

PISA

PISA 2025 Technical Report

Draft version



22

Scaling Procedures and Construct Validation of Context Questionnaire Data

Introduction

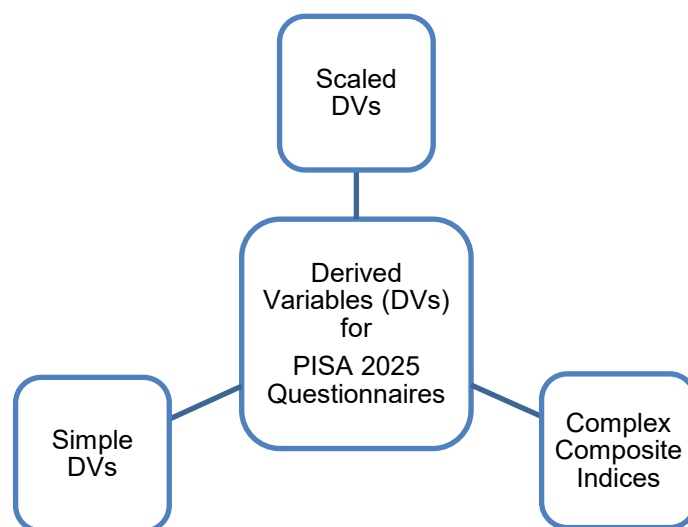
The PISA 2025 context questionnaires were developed in reference to the PISA 2025 questionnaire framework (OECD, 2026) and their development is described in Chapter 6 of this technical report. Many questionnaire items were designed to be combined to represent latent constructs that cannot be observed directly (e.g. a student's science self-efficacy; sense of belonging; or economic, social, and cultural status). To construct meaningful indices, transformations or scaling procedures were applied to these items.

In the following sections, these indices are referred to as *derived variables* (DVs). This chapter describes the DVs that were constructed and validated for all questionnaires administered in PISA 2025 across respondent groups – students, parents, schools, and teachers.

As in previous PISA surveys, three different kinds of DVs can be distinguished (see Figure 22.1):

- *simple questionnaire indices* constructed through the arithmetical transformation or recoding of one or more items
- *scaled indices* based on item response theory (IRT) scaling
- *complex composite indices* based on a combination of two or more indices

Figure 22.1. Types of derived variables for questionnaires in PISA 2025



As described in Chapter 6, the PISA 2025 context questionnaires included a broad scope of contextual factors assessed with different questionnaire instruments. While the student and school questionnaires were implemented in all countries/economies, many countries/economies also administered an optional questionnaire for the parents of the participating students. In addition, countries/economies could choose to administer the optional Information and Communication Technology (ICT) Familiarity Questionnaire responded by students and/or optional Science Teacher Questionnaire, which included questionnaires for teachers of science subjects at the PISA modal grade for 15-year-old students.

In PISA 2025, countries that participated in the Foreign Language Assessment (FLA) also administered core student and school questionnaires and had the option of administering teacher and parent questionnaires. The FLA questionnaires included both selected (general) questions from the core instruments as well as FLA-specific questions. The school FLA questionnaire included all questions in the core school questionnaire and, in addition, a set of FLA-specific questions. Indices that are included in both core and FLA questionnaires are described in the sections about the core instruments, while only the FLA-specific indices are detailed in the sections related to FLA questionnaires.

This chapter describes the methodology used for the scaled DVs and also presents an overview of all the simple and scaled DVs for each questionnaire.

Scaling methodology and reporting of scores

Scaling methodology

As in previous cycles of PISA, many of the DVs were constructed using IRT. More specifically, the two-parameter logistic model (2PLM) (Birnbaum, 1968) was used to scale items with only two response categories (i.e., dichotomous items), while the generalised partial credit model (GPCM) (Muraki, 1992), was used to scale items with more than two response categories (i.e., polytomous items).¹ A detailed explanation of each model is in the following sections. The software ACER Conquest (Adams, Wu, Cloney, Berezner, & Wilson, 2020) (version 5) was used for the scaling.

In the initial scaling, item parameters were estimated using data from all individuals with available data from all participating countries/economies. Each country/economy was included in the analysis using a senate weight (SENWT). The senate weight is a linear transformation of the student full sampling weight (W_FSTUWT) such that the sum of SENWT for all cases within a country/economy add up to a constant of 5,000. Due to missing responses within each country/economy, the sum of the SENWT of the cases used in the calibration of each scale varied on a scale-by-scale basis.

The groups used in the scaling are called *country-by-language groups* since they are defined by both country/economy and language group. For simplicity, the country-by-language groups are also called *groups* in the remainder of this chapter.

In a few cases, scaled item sets included items with negative valence, which means that for these items a higher response category indicated a lower level of the measured construct. The responses to these items were reverse coded prior to scaling. For all items, including the reverse-coded items, the responses were recoded so that the response corresponding to the lowest level of the construct was coded as 0. Any missing response data, whether it was because an item was not administered or a student did not respond to an item, were ignored and were not included in the analysis.

For simplicity, the following sections describe the classical *two-parameter logistic model* (2PLM) and *generalized partial credit model* (GPCM) parameterisations (Birnbaum, 1968; Muraki, 1992), which are mathematically equivalent to the models implemented in ACER ConQuest. In PISA 2025, item calibration was performed using the multidimensional random coefficients multinomial logit model (MRCMLM)

implemented in ACER ConQuest version 5. The relationship between the classical and ConQuest parameterisations is described in Chapter 12.

The two-parameter logistic model (2PLM)

The 2PLM, a generalisation of the Rasch model (Rasch, 1960), assumes that the probability of a response x to be positive (coded as 1 in this case) by individual v to item i depends on the difference between the respondent v 's trait level θ and the location of the item β . In addition, the 2PLM postulates that for every item, the association between this difference and the response probability depends on an additional item discrimination parameter α . The equation for the response probability for an item under the 2PLM is presented in Formula 22.1:

Formula 22.1

$$P_i(x_i = 1 | \theta_n, \alpha_i, \beta_i) = \frac{\exp(D\alpha_i(\theta_n - \beta_i))}{1 + \exp(D\alpha_i(\theta_n - \beta_i))}$$

The item location parameter β_i can be regarded as the item's general location on the latent continuum of the construct being measured. Items with a higher β_i parameter require a higher latent trait for a positive response to be selected.

The item discrimination parameter α , which was scaled by a constant $D = 1.7$ starting in PISA 2015 when the 2PLM was used instead of the Rasch model, characterises how quickly the probability of responding positively to an item approaches 1 with an increase in the trait level θ . In other words, α describes how well a certain item relates to the latent trait θ and, therefore, discriminates between individuals with different trait levels. To solve the indeterminacy of the IRT scale, the average of the person parameters θ across all the pooled sample was constrained to 0.

The generalised partial credit model (GPCM)

The GPCM (Muraki, 1992) is a mathematical model for the probability that an individual will select a certain response category for an item with more than two response categories. Note that the GPCM is a generalisation of the 2PLM and that it reduces to the 2PLM when applied to items with only two response categories. For an item i with $m + 1$ ordered categories, the probability of an individual selecting a certain response category k ($0, 1, 2, \dots, m$) under the GPCM and adopting the same notation employed above can be written as:

Formula 22.2

$$P_i(x_i | \theta_n, \alpha_i, \beta_i, d_{ir}) = \frac{\exp \sum_{r=0}^k D\alpha_i(\theta_n - \beta_i + d_{ir})}{\sum_{u=0}^{m_i} \exp \sum_{r=0}^u D\alpha_i(\theta_n - \beta_i + d_{ir})}, x_i = 1, 2, \dots, m_i$$

As with the 2PLM, the overall item location parameter β_i represent the item i 's general location on the latent continuum. Items with a higher β_i parameter require higher level of the latent trait before higher response categories become more likely. The step parameters d_{ir} (of which there are m for an item with $m + 1$ categories, and which sum to zero within each item) represent the deviations of the category intersections δ from the general location β .

The category intersection δ is the intersection between two neighbouring category characteristic curves, in other words, the point on the latent continuum θ at which a higher response category is more likely to be

selected (e.g., when the individual is more likely to select “disagree” than “strongly disagree”). Note that β and d can be used to calculate the category intersection δ using Formula 22.3.

Formula 22.3

$$\delta_{ir} = \beta_i - d_{ir}$$

The discrimination parameter α , which was scaled by a constant $D = 1.7$ starting in PISA 2015, represents the slope of the category characteristic curves. In other words, it indicates how well selecting a certain response category discriminates between individuals on the latent continuum θ . To solve the indeterminacy of the IRT scale, the weighted mean of the latent trait across the pooled sample was constrained to zero. A special case of the GPCM is when $\alpha = 1$ for all items, in which case the model is equivalent to the partial credit model (PCM) (Masters, 1982).

Figure 22.2 displays the category characteristic curves of a four-category item (e.g., a Likert-type item with response categories “strongly disagree”, “disagree”, “agree”, and “strongly agree”), with the three item parameters used in the GPCM (i.e., α , β and d) represented in the figure. For comparison, Figure 22.2 displays the category characteristic curves of an item for which only the α parameter has been increased while the β and d parameters were kept the same as in Figure 22.2.

Figure 22.2. Category characteristic curves for a four-category item under the generalised partial credit model (GPCM)

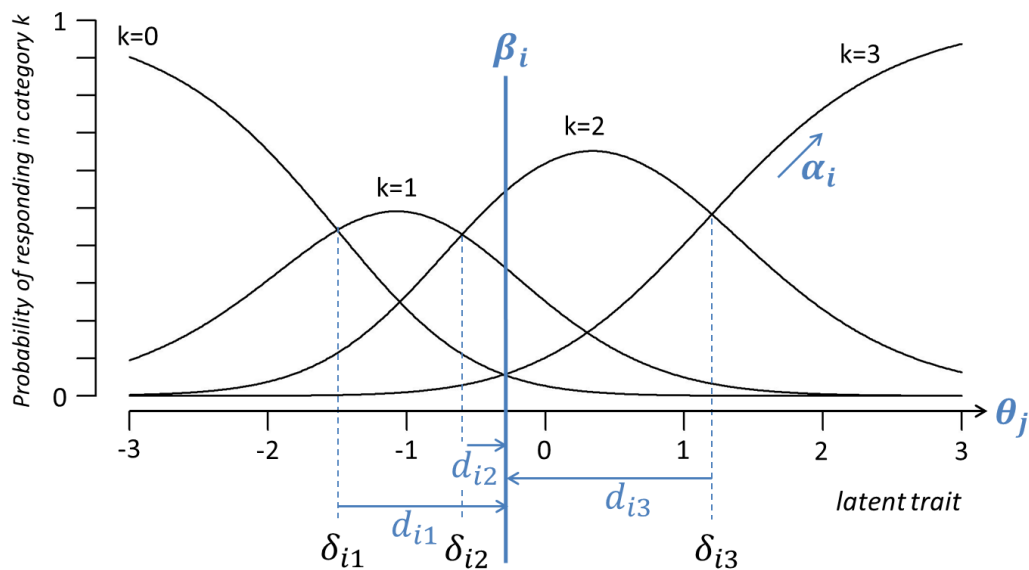
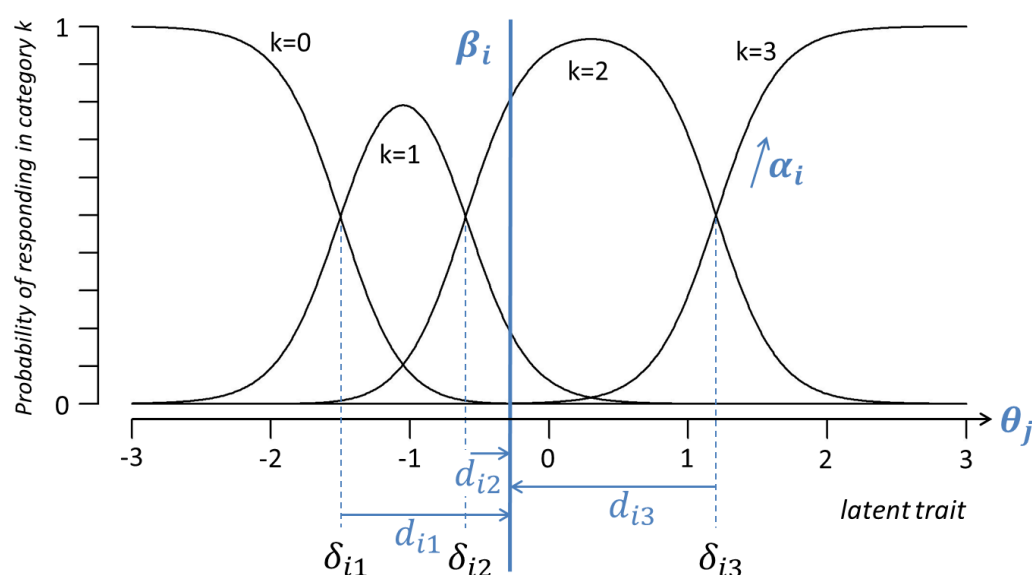


Figure 22.3. Illustration of how an increase in the slope parameter α affects the category characteristic curves of the model above



Special handling of trend scales

For the trend scales, the scaling process began by fixing the item parameters of the trend items to the parameters that had been estimated for each group in the previous cycle, a procedure called fixed parameter linking. The trend scales linked to PISA 2015, 2018 or 2022 were all scaled using the 2PLM and GPCM. To compute trends, a scale needed to have at least three trend items, but some trend scales consisted of both trend items and new items. In this case, the item parameters for the trend items were anchored for the original calibration, while the item parameters for the new items were estimated using the PISA 2025 data. Note that all the items in the trend scales were also evaluated for the goodness-of-fit of the trend parameters, a process described below. Please see Annex Table 22.A.1 for a full list of trend scales included in PISA 2025.

In the previous three PISA cycles (2015, 2018, and 2022), IRT models described above were used with a scaling constant ($D = 1.7$) applied to the discrimination parameter α_i (see Chapter 12). To ensure comparability across cycles, the trend item parameters were transformed to the PISA 2025 parameterization before being used for fixed-parameter linking. Annex Table 22.A.158 reports both parametrisations of item parameters for all trend items included in the PISA 2025 main survey.

Releasing item parameters

PISA 2025 adopted and further refined the approach for evaluating the invariance of latent constructs across groups using multiple-group concurrent calibration with partial invariance constraints, a method which was first introduced in PISA 2015.

As explained above, in the initial scaling, item parameters were estimated using data from all respondents to an instrument with available data from all countries/economies, using weights that ensured approximately equal contributions from each participant. The item parameters that were estimated in this initial scaling process are called *international parameters* since they were estimated using responses from most or all participating countries/economies. After the initial scaling, the fit of the international parameters

for each item was evaluated for each group using the root mean square deviance (RMSD). The $RMSD_g$ for group g is defined as:

Formula 22.4

$$RMSD_g = \sqrt{\int [p_{gobs}(\theta) - p_{gexp}(\theta)]^2 f_g(\theta) d\theta}$$

quantifying the difference between the observed item characteristic curve (ICC) for the group based on the pseudo counts from the E-step of the Expectation-Maximisation (EM) algorithm ($p_{obs,gk}(\theta)$) with the model-based Item Characteristic Curve ICC ($p_{exp,gk}(\theta)$) (Adams, Wu, Cloney, Berezner, & Wilson, 2020). RMSD values range from 0 to 1, with values close to 0 indicating good item fit, meaning that the model-based item parameters fit the data for the group well. Note that the RMSD statistic is sensitive to group-specific deviations of both the item location parameter β and the item discrimination parameter α .

When the RMSD for an item*group exceeded a pre-defined cut-off of 0.25, it was considered that the model-based item parameters did not fit the group's data well, and unique item parameters (also called *group-specific item parameters*) were estimated for the group using data only from that group. However, if more than one group had similar response patterns for an item, data from those groups were pooled together and the same unique parameters were estimated for those groups. This process is called *releasing item parameters*. Item parameters were released until all item*groups had an RMSD value under 0.25. In PISA 2015 and PISA 2018, an RMSD value of 0.3 was used as the cut-off criterion for releasing item parameters for the context questionnaires. For PISA 2022 this threshold was lowered to 0.25 and PISA 2025 continued to apply this threshold.

The final distribution of the RMSD values across groups for each scale item after the final scaling is documented in Annex E. Note that the figures do not include RMSD values for item*groups with an unweighted sample size of 500 or less, as the sample size was considered as too small to estimate stable group-specific ICCs. The minimum threshold for sample size was increased from 150 in PISA 2022 to 500 in PISA 2025 following consultation with the PISA 2025 Technical Advisory Group (TAG). As a consequence, in PISA 2025 no group item releasing was conducted for school questionnaire data due to the relatively low numbers of schools at the group level.

Scale scores

The scaling process described above produced weighted likelihood estimates (WLE) (Warm, 1989) for each individual. These WLE scores were subsequently standardised through the process described in the following sections. Note that if an individual had fewer than three valid responses for a scale, a WLE score was not produced for the individual and his/her scale score was replaced with "99" in the SPSS file and ".M" in the SAS file.

New scales

For the new scales, the original WLE scores were transformed into a reporting metric to have a mean of 0 and a standard deviation of 1 across the OECD countries, using senate weights for all cases with available data. The transformation was achieved by applying Formula 22.5:

Formula 22.5

$$\theta'_v = \frac{\theta_v - \bar{\theta}_{OECD}}{\sigma_{\theta(OECD)}}$$

where θ'_v is the scale score on the reporting metric, θ_v is the original WLE, $\bar{\theta}_{\text{OECD}}$ is the mean of the original WLEs across the OECD countries, and $\sigma_{\theta \text{OECD}}$ is the standard deviation of the original WLEs across the OECD countries. The transformation constants that were used to transform the original WLEs into the reporting scale are displayed in Annex Table 22.A.2.

For the new scales, an average scale score of 0 is expected when calculated across all OECD countries using the senate weights. A negative scale score does not imply that a respondent replied negatively to the items in the scale. Rather, it means that a respondent has a score that is below the OECD average.

Trend scales

For the trend scales, to ensure the comparability of the scale scores from the current cycle to the scale scores from the previous cycle, the original WLEs of PISA 2025 were transformed using the same transformation constants of the original WLEs from the cycle to which the current cycle was linked. Annex Table 22.A.3, Annex Table 22.A.4, and Annex Table 22.A.5 present, respectively, the OECD means and standard deviations of the original WLEs in PISA 2022, 2018 or 2015 for the trend scales linked to these previous cycles.

For some of the PISA 2015 scales, that were linked to PISA 2006, a different transformation was used (for a technical description, see OECD, 2017, p. 292f), which was applied as follows:

$$\theta'_v = A * \theta_v + B$$

The same linear transformations (A and B) were applied as in PISA 2015¹ and are presented in Annex Table 22.A.6.

Criteria for suppressing scale scores

The scale scores of individuals or groups were suppressed under the following conditions.

Low internal consistency

Cronbach's alpha coefficient was used to check the internal consistency of each scale for each group. This coefficient ranges from 0 to 1, with a higher value indicating higher internal consistency. A group needed to have a Cronbach's alpha of at least 0.60 for a scale in order for the group's scale scores to be reported. Scale scores were suppressed for countries/economies (or adjudicated regions with separate national centres) in which one or more language groups had a Cronbach's alpha under 0.60 for the scale. Only language groups with sample sizes of at least 40 cases and represented 25 per cent of total weighted sample in a country/economy were considered for a suppression of scale scores.

Few items with international parameters

For each scale, a group needed to have at least three items with international parameters in order for the scale scores of the group to be considered comparable to the scale scores of the other groups. Scale scores were suppressed for countries/economies in which one or more language groups had less than three items with international parameters for the scale. The scale scores for the individuals in these countries/economies were replaced with "97" in the SPSS file.

¹ Please note that there is an error in Table 16.2 of the PISA 2015 technical report (OECD, 2017) as for the respective linear transformation the headings for the columns for A and B were swapped (and presented as B and A).

Lack of trend items with international parameters

For the trend scales, a group needed to have at least three trend items with international parameters in order for the PISA 2025 scales scores for the group to be considered comparable to the scale scores of the previous cycle to which the current cycle was linked. Scale scores were suppressed for countries/economies in which one or more language groups had less than three trend items with international parameters for the scale. The scale scores for the individuals in these countries/economies were replaced with “97” in the SPSS file.

Student Questionnaire derived variables

There were 55 indices that were derived from the student questionnaire, including 28 simple DVs, 26 IRT-scaled DVs, and one complex composite index (ESCS). The DVs are shown in Annex Table 22.A.7 and are described in the following section. The first section covers all simple DVs, the second section covers those that are based on IRT scaling, and the last section covers the complex composite index. The simple and scaled DVs are organised first by framework module (please see Chapter 6 for a description of modules) and then in alphabetical order within modules.

Simple questionnaire indices

Basic demographics (Module 1)

Student's age (AGE)

The age of a student (AGE) was calculated as the difference between the year and month of the testing and the year and month of a student's birth, which was obtained from school records from the student sampling data and validated by comparing to the students' responses in the questionnaire. Data on students' age were obtained from both the questionnaire (ST003) and the student tracking forms. The formula for computing AGE was:

Formula 22.6

$$AGE = \frac{12(Ty - Sy) + (Tm - Sm)}{12}$$

where Ty and Sy are the year of the test and the year of the students' birth, respectively, in four-digit format (for example, “2025” or “2010”), and Tm and Sm are the month of the test and month of the students' birth, respectively. The result is rounded to two decimal places.

Grade compared to modal grade in country (GRADE)

The relative grade index (GRADE) was computed to capture between-country/economy variation. It indicates whether students are in the country/economy's modal grade (value of 0), or the number of grades below or above the modal grade in the country. The information about the students' grade level was obtained from school records from the student sampling data and validated by comparing the students' responses in the Student Questionnaire (ST001).

Gender (ST004D01T)

The gender of a student which was obtained from school records from the student sampling data and validated by comparing to the student's responses in the questionnaire (ST430). ST004D01T takes the value 1 for female students and 2 for male students. The index was assigned a missing value in countries and economies which included additional national categories in the sampling records.

Male gender (MALE)

Index derived from ST004D01T. The index takes the value 1 for students classified as male in ST004D01T and the value of 0 for students classified as female or any other national category included in the student sampling records. If student gender information is missing from the questionnaire or the sampling information, this index was assigned a missing value.

Economic, social and cultural status (Module 2)

Parent/guardian 1's level of education (P1ISCED)

Student responses to question ST405 or ST410_A regarding their parent/guardian 1's education were used to derive the parent 1's level of education (P1ISCED) index, where education level ranged from "1" none of the listed education ISCED levels to "9" ISCED level 8, as noted in Annex Table 22.A.8.²

Parent/guardian 2's level of education (P2ISCED)

Student responses to question ST410_B regarding their parent/guardian 2's education were used to derive the parent 2's level of education (P2ISCED) index, where education level ranged from "1" none of the listed education ISCED levels to "9" ISCED level 8, as noted in Annex Table 22.A. 8.

Highest level of education of parents/guardians (HISCED)

Students' responses to questions ST405 or ST410 regarding their parents/guardians 1 and 2's education were used to derive the index of highest education level of parents (HISCED). The index is equal to the highest ISCED level of either parent/guardian.

Highest education of parents/guardians in years (PAREDINT)

The index of the highest education of parents/guardians in years, PAREDINT, was based on the median cumulative years of education associated with completion of the highest level of education of parents/guardians (HISCED). Cumulative years of education values used in PISA 2018 were assigned to each ISCED level (see Annex Table 22.A.8).

Parent/guardian's occupational code (OCOP1 and OCOP2)

Students' responses to the fill-in question ST408 or ST491 about their parent/guardian 1's occupation and question ST492 about their parent/guardian 2's occupation were human-coded based on the International Standard Classification of Occupations (ISCO)-08 classification system, resulting in the occupational code of 4-digits (ILO, 2007). These 4-digit codes were reduced to 2-digit codes for public use data files. The 2-digit codes in OCOP1 and OCOP2 indices range from 0 to 97. Codes 00 to 96 are occupations from the ISCO-08 classification system. Codes 97 were used to classify responses that fell outside of the ISCO-08 classification system – for example, unemployed, retired, sick or vague responses.

Parent/guardian's occupational status (BP1MJ and BP2MJ)

The parent/guardian 1 and parent/guardian 2 occupational status index, BMMJ1 and BFMJ2, were derived from the 4-digit ISCO coded responses in OCOP1 and OCOP2 (before reduction to 2-digits for public use data files) indices, respectively, and international socio-economic index of occupational status (ISEI) (Ganzeboom & Treiman, 2003) scores. The 4-digit ISCO-08 occupation codes were mapped onto ISEI ratings.

Highest occupational status of parents/guardians (HISEI)

The index of the highest occupational status of parents/guardians (HISEI) was based on the 4-digit ISCO-08 occupational codes that were human coded from students' responses to questions ST408 or ST491 and ST492 about their parent/guardian 1 and parent/guardian 2's occupations, respectively. The index was equal to the highest of their parent/guardian 1's (BP1MJ) and parent/guardian 2's (BP2MJ) ISEI scores.

M3 – Educational career (Module 3)

Study programme level and orientation (ISCEDP)

PISA collects data on study programmes available to 15-year-old students in each country/economy. This information is obtained through the student tracking form and the Student Questionnaire (ST002). All study programmes were classified using the International Standard Classification of Education (ISCED 2011).

The study programme level and orientation index (ISCEDP) is a three-digit index that describes whether students were at the lower or upper secondary level and (ISCED 2 or ISCED 3) and whether their programmes were general or vocational and sufficient for level completion with direct access to tertiary or post-secondary non-tertiary education. ISCEDP values and labels can be found in Annex Table 22.A.9.

In the final database, all national programmes are included in a separate DV (PROGN) where the first six digits represent the National Centre code, and the last two digits are the nationally specific programme code.

Grade repetition (REPEAT)

Students' answers on question ST127 of whether and, if yes, how often they have ever repeated a grade at ISCED levels 1, 2, and 3 were combined into the index REPEAT. Each ISCED level was represented by one item and included three response options ("No, never", "Yes, once", "Yes, twice or more"). REPEAT took the value of "0" if the student never repeated a grade (student did not select options 2 or 3 for any of the three items) and the value of "1" if the student repeated a grade at least once (student selected options 2 or 3 for at least one of the three items). The index was assigned a missing value if none of the three response options were selected in any levels.

Skipping classes or days of school (SKIPPING)

Students' responses to whether, in the two weeks prior to the PISA test, they had skipped classes (ST062Q02TA) or days of school (ST062Q01TA) at least once were used to derive an indicator of student truancy. Both questions have four response options ("Never", "One or two times", "Three or four times", "Five or more times"). The indicator takes a value of 0 if students reported that they had not skipped any class or day of school in the two weeks before the PISA test, and a value of 1 if students reported that they had skipped classes or days of school at least once in the same period.

Arriving late for school (TARDYSD)

Students responded to a question about whether and how frequently they had arrived late for school during the two weeks prior to the PISA test (ST062Q03TA). TARDYSD takes a value of “0” for on-time students if students reported that they had not arrived late for school, a value of “1” for occasional late arrivals if students report they arrived late for school one or two times, and “2” for frequent late arrivals if students reported they had arrived late for school three or more times.

Migration and language exposure (Module 4)

Based on students’ responses to question ST413 or ST414 (“In what country were you and your parents born?”), four indices are created as outlined below.

Student’s born in country of test (BICT_S)

This index has the value “1” if the student selected the country of test (“Country A”) in question ST413AQ01D or in ST414AQ01D, and “0” otherwise. The variable COBN_S contains the country/economy-specific code for each country students selected in ST413AQ01D or ST414AQ01D.

Parents/guardians born in country of test (BICT_P1 and BICT_P2)

BICT_P1 has the value “1” if the student selected the country of test (“Country A”) in question ST413BQ01D or in ST414BQ01D, and “0” otherwise. BICT_P2 has the value “1” if the student selected the country of test (“Country A”) in question ST414CQ01D, and “0” otherwise.

The variables COBN_P1 and COBN_P2 contain the country/economy-specific code for each country students selected in ST413BQ01D or ST414BQ01D and ST414CQ01D if applicable.

Index on immigrant background (IMMIG)

The index on immigrant background (IMMIG) is calculated from the three variables above (BICT_S, BICT_P1, BICT_P2), and has the categories as listed below. Students with missing responses for either the student or for both parents were given missing values for this variable.

1. Native students (those students who had at least one parent born in the country/economy);
2. Second-generation students (those born in the country/economy of assessment but whose parent[s] were born in another country/economy);
3. First-generation students (those students born outside the country/economy of assessment and whose parents were also born in another country/economy).

Language spoken at home (LANGN)

Students also indicated what language they usually spoke at home (ST022), and the database includes a variable (LANGN) containing country/economy-specific code for each language.

Post-secondary preparedness and expectations (Module 9)

Highest expected educational level (EXPECEDU)

Students’ responses which of a list of possible educational levels they expect to complete in question ST327 were transformed into the index of “Highest Expected Educational Level”. Values on the index can range from “Less than ISCED level 2” to “ISCED level 8”. Scores are assigned as shown in Annex Table 22.A.10.

Expected occupation (OCOD3) and Expected occupation status (BSMJ)

Students' responses to the fill-in question ST329 about what kind of job they expect to have when they are about 30 years old were human-coded based the ISCO-08 classification system, resulting in the index "Expected Occupation (OCOD3)". These ISCO codes were then mapped to the international socio-economic index of occupational status (ISEI) (Ganzeboom & Treiman, 2003) in variable BSMJ. Higher scores on this variable indicate higher levels of a student's expected occupational status.

Clear idea about future job (SISCO)

The students who had a clear idea about their future job index (SISCO) was based on the human-coded open-ended expected occupation index, OCOD3, which was derived from question ST329. Students who had no clear idea about their future jobs were considered those who indicated "I do not know" or gave a vague answer such as "a good job", "a quiet job", "a well-paid job", "an office job" in response to question ST329. In the OCOD3 index, "I don't know" responses were coded 9704 and vague responses were coded 9705. Examples of invalid responses include students who did not answer the question or gave an answer, such as a smiley face. Specifically, a value of "0" is assigned on the index if OCOD3 values are 9704 or 9705, and a value of "1" is assigned if OCOD3 values are 0000 to 9703.

Experiences out of regular school hours (Module 10)

Working for pay before or after school (WORKPAY)

Students' answers on how many days during a typical school week they worked for pay before going to school and/or after leaving school in questions ST294 and ST295 were recoded into the index of "Work for pay before or after school". Each item included six response options ("0 days", "1 day", "2 days", "3 days", "4 days", "5 or more days"). Values on this index range from 0 (no work for pay) to 10 (10 or more times of working for pay per week).

Exercise or practise sport (EXERPRAC)

Students' answers on how many days during a typical school week they Exercise or practise a sport before going to school and/or after leaving school in questions ST294 and ST295 were recoded into the index of "Exercise or practise sport". Each item included six response options ("0 days", "1 day", "2 days", "3 days", "4 days", "5 or more days"). Values on this index range from 0 (no work in household or care of family members) to 10 (10 or more times of working in household or caring for family members per week).

Derived variables based on IRT scaling

The student questionnaire provided data for 28 DVs based on IRT scaling. The Cronbach's alpha for each scale and group are presented in Annex Table 22.A.11, the number of items with international parameters for each scale and group are presented in Annex Table 22.A.12, the number of trend items with international parameters for each trend scale and group are presented in Annex Table 22.A.13, the countries/economies for which the scale scores were suppressed for each scale are presented in Annex Table 22.A.14, and the groups that did not administer each scale are presented in Annex Table 22.A.15.

Economic, social and cultural status (Module 2)

Home possessions (HOMEPOS)

In the HOMEPOS scale (which included questions ST250, ST251, ST254, and ST255), students indicated whether their household possessed certain items (e.g., "a room of your own", "educational software or apps") – ST250, or how many of an item their household possessed (e.g., "rooms with a <flush toilet>",

“cars, vans, or trucks”) – ST251. In addition, students answered how many different types of digital devices (ST254) and books (ST255) were in their home.

This scale included a maximum of 24 items for each country/economy:

- Five international items in ST250 (ST250Q01JA, ST250Q02JA, ST250Q03JA, ST250Q04JA, ST250Q05JA)
- Up to 6 items subject to selection in ST250 from a list of 20 proposed items (ST250Q08DA to ST250Q28DA).
- Five international items in ST251 (ST251Q01JA, ST251Q03JA, ST251Q04JA, ST251Q06JA, ST251Q07JA)
- Two country/economy-specific items (ST251Q08JA, and ST251Q09JA) that were seen as local measures of family wealth within the country/economy’s context.³
- Five international items in ST254 (ST254Q01JA, ST254Q02JA, ST254Q03JA, ST254Q04JA, ST254Q05JA)
- One international item number of books in the home (ST255Q01JA)

Note that all groups received unique item parameters for the country/economy-specific items in ST251 (i.e., no international parameters were estimated for these items) and that for some items in ST251 and ST255, the response categories were collapsed to align with the response categories used in previous cycles. Annex Table 22.A.16 shows the item wording and item parameters for the items in this scale, while Annex Table 22.A.17 shows how the response categories for each item were recoded prior to scaling. This scale was also derived from the FLA student questionnaire using anchored parameters and set to the same metric.

ICT resources (ICTRES)

Students reported on the availability of 11 Information and Communications Technologies (ICT) resources in their home (e.g., “a computer (laptop, desktop, or tablet) that you can use for school work” , “your own tablet”, “televisions”) in questions ST250 (which had two response categories) and ST254 (which had four substantive response categories and an additional response category “I don’t know.” which was recoded as missing prior to scaling). These items were scaled into the index of “ICT resources”, which is linked to the PISA 2022 scale ICTRES through the anchoring of parameters for 10 out of 11 items. Annex Table 22.A.18 shows the item wording and item parameters for the items in this scale. This index is also included in the FLA student questionnaire. This scale from the FLA student questionnaire was set to the same metric as the scale from the core student questionnaire by using anchored parameters.

School culture and climate (Module 6)

Exposure to Bullying (BULLIED)

Students’ ratings of how often they had a range of experiences at school that are indicative of being bullied during the past 12 months (e.g., “Other students left me out of things on purpose.”, “Other students made fun of me.”) in question ST038 were scaled into the index of “Exposure to Bullying”. Note that, with anchored parameters for four link items, this scale was linked to the scale BULLIED in PISA 2022, which in turn was linked to the scale called BEINGBULLIED PISA 2018. Each of the five items included in this scale had four response options (“Never or almost never”, “A few times a year”, “A few times a month”, “Once a week or more”). Annex Table 22.A.19 shows the item wording and item parameters for the items in this scale. It also indicates the four link items.

Feeling safe (FEELSAFE)

Students' ratings of their agreement with four statements about their perceived safety (e.g., "I feel safe on my way to school.", "I feel safe in my classrooms at school.") in question ST265 were scaled into the index of "feeling safe". Note that this scale is linked to the scale FEELSAFE in PISA 2022 through anchoring parameters for the four link items. Each of the four items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). Annex Table 22.A.20 shows the item wording and item parameters for the items in this scale.

Sense of belonging (BELONG)

Students' ratings of their agreement with six statements (e.g., "I feel like I belong at school.", "I feel lonely at school.") in question ST034 were scaled into the index of "Sense of belonging". Note that through anchoring of item parameters for six link items this scale was linked to the scale BELONG in PISA 2022, which in turn was linked to the scale with the same name in PISA 2018. Each of the six link items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). Annex Table 22.A.21 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling. This scale from the FLA student questionnaire was set to the same metric as the scale from the core student questionnaire by using anchored parameters.

Science identity (Module 7)

Collective environmental self-efficacy (COLLENEFF)

Students' ratings of agreement with different statements related to their beliefs in collective agency to resolve environmental issues (e.g., "I believe that by working with my local community we can play an important role in positively impacting the environment", "I believe that by working with my school we can reduce their energy consumption") in question ST499 were scaled into the new PISA 2025 index of "collective environmental self-efficacy". Each of the four items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.22 shows the item wording and item parameters for the items in this scale.

Enjoyment of science (JOYSCIE)

Students' ratings of agreement with different statements related to their enjoyment of learning about broad science (e.g., "I generally have fun when I am learning <broad science> topics", "I enjoy acquiring new knowledge in <broad science>") in question ST094 were scaled into the index of "enjoyment of science". Note that this scale is linked to the scale JOYSCIE in PISA 2015 based on all items in this question. Each of the four items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.23 shows the item wording and item parameters for the items in this scale.

Opportunities for environmental learning in class (OPENVLRN)

Students' ratings of agreement with different statements related to their perceptions of environmental learning in class (e.g., "my teacher takes us on field trips to investigate environmental issues", "my teacher provides opportunities to research environmental issues") in question ST501 were scaled into the new PISA 2025 index of "opportunities for environmental learning in class". Each of the four items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.24 shows the item wording and item parameters for the items in this scale.

Science self-efficacy (EFFSCIE)

Students' ratings of how confident they felt about having to do a range of science tasks (e.g., "describe the role of antibiotics in the treatment of disease", "identify the reliability of scientific information") in question ST129 were scaled into the new PISA 2025 index of "science self-efficacy". Note that while there were two link items (ST129Q03TA and ST129Q05TA) that were also included in PISA 2015, this scale could not be linked to the scale SCIEEFF in PISA 2015 due to low number of common item material. Each of the six items included in this scale had four response options ("I could do this easily", "I could do this with a bit of effort", "I would struggle to do this on my own", "I couldn't do this") and all of these were reverse-coded so that higher scale scores reflect higher levels of self-efficacy. Annex Table 22.A.25 shows the item wording and item parameters for the items in this scale.

Students' beliefs about the nature of science (STBELSCIE)

Students' ratings of agreement with different statements related to their beliefs about the nature of science (e.g., "good answers are based on evidence from many different experiments", "new discoveries can change what scientists think is true") in question ST131 were scaled into the new PISA 2025 index of "students' beliefs about the nature of science". Note that while the first item this scale is identical with an item used in PISA 2015, the scale is not linked to the scale EPIST derived in PISA 2015. Each of the five items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.26 shows the item wording and item parameters for the items in this scale.

Students' beliefs in their capacity to generate environmental change (ENVCAPCH)

Students' ratings of agreement with different statements related to their beliefs knowing how to resolve environmental issues (e.g., "I know how to work with others to positively impact the environment", "If people work together, we can resolve environmental issues") in question ST500 were scaled into the new PISA 2025 index of "students' beliefs in their capacity to generate environmental change". Each of the four items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.27 shows the item wording and item parameters for the items in this scale.

Students' environmental awareness (ENVAWARE)

Students' ratings of how informed they felt about a range of environmental issues (e.g., "the increase of greenhouse gases in the atmosphere", "extinction of plants and animals") in question ST092 were scaled into the index of "environmental awareness". Note that, through anchoring parameters for four common items (ST092Q01TA, ST092Q04TA, ST092Q05TA, ST092Q08NA), this scale is linked to PISA 2015. Each of the six items included in this scale had four response options ("I have never heard of this", "I have heard about this but I would not be able to explain what it is really about", "I know something about this and could explain the general issue", "I am familiar with this and I would be able to explain this well"). Annex Table 22.A.28 shows the item wording and item parameters for the items in this scale.

Students' participation in environment-related activities (ENVAPART)

Students indicated the frequency of being involved in environment-related activities (e.g., "joining a peaceful march or rally related to an environmental issue", "making purchases based on how sustainable the products are") in question ST495 were scaled into the new PISA 2025 index of "students' participation in environment-related activities". Each of the four items included in this scale had four response options ("never", "once", "a few times", "many times"). Annex Table 22.A.29 shows the item wording and item parameters for the items in this scale.

General social and emotional characteristics (Module 8)

Curiosity (CURIO)

Students' ratings of their agreement with statements about a range of behaviours indicative of curiosity (e.g., "I like to know how things work.", "I am more curious than most people I know.") in question ST301 were scaled into the new PISA 2025 index of "curiosity". Note that one negatively worded item showed unsatisfactory measurement properties in PISA 2025 and was not used for scaling. As only four out of 10 items used in PISA 2022 were available for scaling, this scale is not linked to the scale CURIOVAGR in PISA 2022. Each of the four items included in this scale had five response options ("Strongly disagree", "Disagree", "Neither agree nor disagree", "Agree", "Strongly agree"). Annex Table 22.A.30 shows the item wording and item parameters for the items in this scale.

Perseverance (PERSEV)

Students' ratings of their agreement with statements about a range of behaviours indicative of perseverance (e.g., "I keep working on a task until it is finished.", "I apply additional effort when work becomes challenging") in question ST307 were scaled into the new PISA 2025 index of "Perseverance". Note that two negatively worded items showed unsatisfactory measurement properties in PISA 2025 and were not used for scaling. As only four out of 10 items used in PISA 2022 were available for scaling, this scale is not linked to the scale PERSEVAGR in PISA 2022. All four items included in this scale had five response options ("Strongly disagree", "Disagree", "Neither agree nor disagree", "Agree", "Strongly agree"). Annex Table 22.A.31 shows the item wording and item parameters for the items in this scale. This scale from the FLA student questionnaire was set to the same metric as the scale from the core student questionnaire by using anchored parameters.

Exposure to science content (Module 15)

Engagement with science practices (ENGSCIPR)

Students' frequency ratings of how often they do different science practices in science courses (e.g., "develop models or simulations to help explain natural phenomena e.g. earthquakes", "develop scientific arguments using evidence to support a claim") in question ST515 were scaled into the new PISA 2025 index "engagement with science practices". Each of the six items included in this scale had four response options ("Every lesson", "most lessons", "some lessons", "never or almost never") and all items were reverse-coded prior to scaling. Annex Table 22.A.32 shows the item wording and item parameters for the items in this scale.

Science teacher behaviours (Module 16)

Cognitive activation in science (COGACSC)

Students' frequency ratings of how often their science teacher showed a range of behaviours indicative of fostering science reasoning during the ongoing school year (e.g., "the teacher asked us to think about how new and old scientific topics were related", "the teacher taught us to memorise scientific facts") in question ST469 were scaled into the new PISA 2025 index of "cognitive activation in broad science". Each of the five items included in this scale had five response options ("never or almost never", "less than half of the lessons", "about half of the lessons", "more than half of the lessons", "every lesson or almost every lesson"). Annex Table 22.A.33 shows the item wording and item parameters for the items in this scale.

Collaborative reasoning in science classes (SCIECOLLR)

Students' frequency ratings of how often their science teacher showed a range of behaviours indicative of the teachers' collaborative reasoning during the ongoing school year (e.g., "the teacher asked us to debate scientific topics with other members of the class", "the teacher asked us to explain a scientific principle to the class") in question ST468 were scaled into the new PISA 2025 index of "collaborative reasoning in science classes". Each of the four items included in this scale had five response options ("never or almost never", "less than half of the lessons", "about half of the lessons", "more than half of the lessons", "every lesson or almost every lesson"). Annex Table 22.A.34 shows the item wording and item parameters for the items in this scale.

Disciplinary climate in science classes (DISCLISCI)

Students' frequency ratings of how often a range of situations occurred in their science lessons (e.g., "Students do not listen to what the teacher said.", "Students get distracted by using <digital resources> (e.g., smartphones, websites, apps).") in question ST097 were scaled into the index of "disciplinary climate in science". Note that, by anchoring parameters for five trend items, this scale was linked to the DISCLISCI scale in PISA 2015. Each of the six items included in this scale had four response options ("every lesson", "most lessons", "some lessons", "never or almost never") and was reverse-coded prior to scaling. Annex Table 22.A.35 shows the item wording and item parameters for the items in this scale, and it also indicates which items are trend items.

Science teacher support (TEACHSUP)

Students' frequency ratings of how often a range of situations occurred in their science lessons (e.g., "The teacher shows an interest in every student's learning.", "The teacher gives extra help when students need it.") in question ST100 were scaled into the index of "science teacher support". Note that this scale was linked using all five items to the TEACHSUP scale in PISA 2015. Each of the five items included in this scale had four response options ("every lesson", "most lessons", "some lessons", "never or almost never"). Annex Table 22.A.36 shows the item wording and item parameters for the items in this scale.

Parental/guardian involvement and support (Module 19)

Perceptions of family support (FAMSUP)

Students indicated the frequency of perceiving family support for learning (e.g., "discuss how well you are doing at school", "take an interest in what you are learning at school") in question ST300 were scaled into the index of "perceptions of family support". Note that this scale is linked through the anchoring of parameters for all five items to the PISA 2022 scale FAMSUP. Each of the five items included in this scale had five response options ("never or almost never", "about once or twice a year", "about once or twice a month", "about once or twice a week", "every day or almost every day"). Annex Table 22.A.37 shows the item wording and item parameters for the items in this scale. This scale from the FLA student questionnaire was set to the same metric as the scale from the core student questionnaire by using anchored parameters.

Learning in the digital world (LDW)

AI use at school (AIUSESCH)

Students indicated the frequency of using AI for different purposes at school (e.g., "to summarize a text I had to read", "to conduct preliminary research on a new topic") in question ST438 were scaled into the new PISA 2025 index of "AI use at school". Each of the four items included in this scale had five response options ("never or almost never", "about once or twice a year", "about once or twice a month", "about once

or twice a week”, “every day or almost every day”). Annex Table 22.A.38 shows the item wording and item parameters for the items in this scale.

Cognitive adaptability (COGABIL)

Students indicated how well different statements described them (e.g., “I can handle unusual situations”, “I can adapt easily to a new culture”) in question ST216 were scaled into the new PISA 2025 index of “cognitive adaptability”. Each of the four items included in this scale had five response options (“very much like me”, “mostly like me”, “somewhat like me”, “not much like me”, “not at all like me”) and all items were reverse-coded prior to scaling. Annex Table 22.A.39 shows the item wording and item parameters for the items in this scale.

Enjoyment of problem solving (ENPROBS)

Students’ ratings of agreement with different statements related to their perceptions of problem solving (e.g., “I enjoy finding solutions to problems”, “I keep working on a problem even if I might not be able to solve it”) in question ST435 were scaled into the new PISA 2025 index “enjoyment of problem solving”. Each of the four items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.40 shows the item wording and item parameters for the items in this scale.

Goal setting (GOALSET)

Students indicated the frequency of using different ways of learning for school (e.g., “I plan my approach to studying”, “I ask myself questions about the material to check if I have understood everything”) in question ST432 were scaled into the new PISA 2025 index of “goal setting”. Each of the four items included in this scale had five response options (“never or almost never”, “sometimes”, “often”, “very often”). Annex Table 22.A.41 shows the item wording and item parameters for the items in this scale.

Self-regulation (SELFREG)

Students’ ratings of agreement with different statements about their self-perceptions as a learner (e.g., “when I do not fully understand something, I ask my classmates”, “I only ask for help after trying hard to find the solution on my own”) in question ST433 were scaled into the new PISA 2025 index of “self-regulation”. Each of the four items included in this scale had four response options (“strongly agree”, “agree”, “disagree”, “strongly disagree”) and items were reverse-coded prior to scaling. Annex Table 22.A.42 shows the item wording and item parameters for the items in this scale.

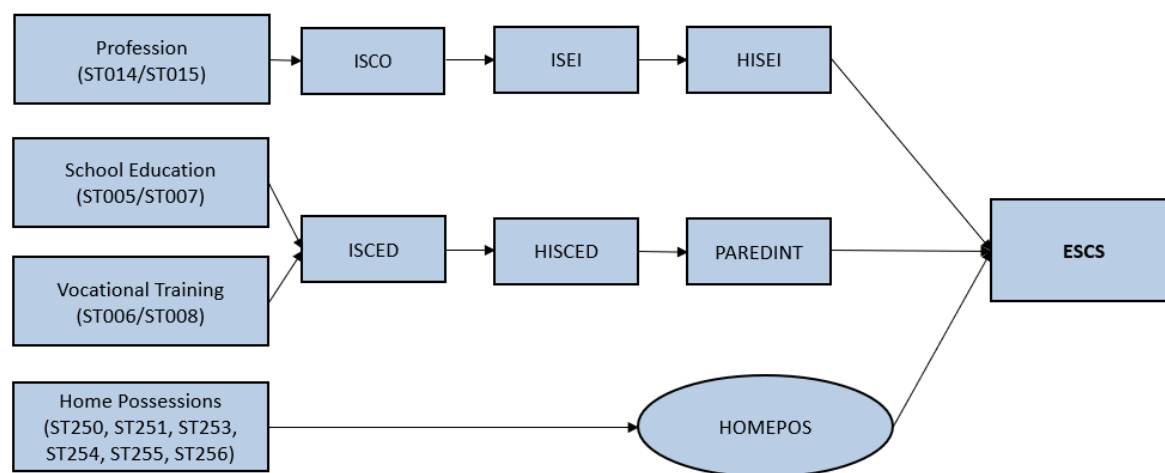
Complex composite index – Index of economic, social and cultural status (ESCS)

There was only one complex composite index derived from the Student Questionnaire – the index of economic, social and cultural status (ESCS).

Components of ESCS

The ESCS score was based on three indicators: highest parental occupation status (HISEI), highest education of parents in years (PAREDINT), and home possessions (HOMEPOS). The rationale for using these three components, which are consistent with the components used in previous PISA cycles, is that socio-economic status is most commonly theoretically conceptualized based on “the big 3” (occupational status, education, and income) (Cowan, et al., 2012). As no direct income measure is available in the PISA data, the existence of household items has been used as a proxy for family income. Figure 22.4 provides a schematic representation of ESCS and its components.

Figure 22.4. Computation of ESCS in PISA 2022



HISEI. For more information on HISEI, refer to the explanation on HISEI in the simple indices section above.

PAREDINT. For more information on PAREDINT, refer to the explanation on PAREDINT in the simple indices section above.

HOMEPOS. For more information on HOMEPOS, refer to the explanation on HOMEPOS in the IRT scale section above.

Computation of ESCS

The ESCS scores were computed using the same methodology used in PISA 2022 (Avvisati, 2020; OECD, 2020; OECD, 2024). For students with missing data on one out of the three components, the missing component was imputed using a regression equation which was created for each country/economy using data from students without any missing components. For each student with a missing component, this regression equation was used to predict the missing component with the two non-missing components and a random value was added to the predicted value to reflect the error of the regression model.⁴ If a student had missing data on more than one component, the ESCS score was not computed for the student, and the student's ESCS score was replaced with "99" in the SPSS file.

After the imputation process, each of the three components (including the imputed values) was standardised to have a mean of 0 and a standard deviation of 1 across the OECD countries, with each OECD country weighted approximately equally using senate weights.⁵ The OECD means and standard deviations that were used to standardise each component of ESCS are displayed in Annex Table 22.A.43.

Subsequently, the arithmetic mean of the three standardised components was calculated to create a preliminary ESCS score for each student. Lastly, the preliminary ESCS scores were standardised again to have a mean of 0 and a standard deviation of 1 across the OECD countries (again with each country weighted approximately equally using senate weights⁶) as the final ESCS score for each student. The OECD mean and standard deviation that were used to transform the preliminary ESCS scores into the final ESCS scores are also displayed in Annex Table 22.A.43.

In contrast to what was done in previous cycles, the ESCS computed for PISA 2025 is not treated as a trend scale. This is due to considerable changes to the section in the student questionnaire in terms of how parental education, occupation and home possessions were captured.

The ESCS index was also derived from FLA Student Questionnaire data using the same procedures as described in this section, for which the metric was set using the same transformations as for the core instrument.

ICT Familiarity Questionnaire indices

The ICT Familiarity Questionnaire is an international option that countries/economies could choose to implement. It was administered to students after they had completed the Student Questionnaire. There were 13 variables derived from this questionnaire, including one simple DVs and 12 IRT scaled DVs. Except for one new IRT scale, all IRT scales were linked to previous PISA cycles (nine to PISA 2022 and two to PISA 2018). However, for some trend scales PISA 2025 employed additional items, corresponding to changes in the ICT familiarity framework. An overview of all DVs in this questionnaire is shown in Annex Table 22.A.44 and each are described in the following sections.

Simple questionnaire indices

Availability and usage of ICT at school (ICTAVSCH)

The availability of ICT at school was gathered from IC170 where students' frequency ratings of how often they use various digital resources at school (e.g., "Desktop or laptop computer", "Smartphone") was used for the index of "ICT availability at school". Each of the seven items in this question included six response options ("Never or almost never", "About once or twice a month", "About once or twice a week", "Every day or almost every day", "Several times a day", "This resource is not available to me at school"). The index was calculated as the number of all seven items that were marked with a value other than "This resource is not available to me at school", thus ranging from 0-7. All items were included in the PISA 2025 ICT Questionnaire.

Availability and usage of ICT outside of school (ICTAVHOM)

The availability of ICT outside of school was gathered from IC171 where students' frequency ratings of how often they use various digital resources outside of school (e.g., "Desktop or laptop computer", "Smartphone") was used for the index of "ICT use outside of school". Each of the six items in this question included six response options ("Never or almost never", "About once or twice a month", "About once or twice a week", "Every day or almost every day", "Several times a day", "This resource is not available to me outside of school"). For each of the six items, a score of "0" was assigned when students choose the "This resource is not available to me outside of school" response options and all other responses were coded "1". The index was calculated as the sum of "0" and "1" designations across the six items that were marked with a value other than "This resource is not available to me at school", thus ranging from 0-6. All items were included in the PISA 2022 ICT Questionnaire.

Distress from online content and cyberbullying (ICTDISTR)

Students' ratings of how upset they were when various situation occurred online (e.g., "Encountering content online that was inappropriate for my age", "Receiving unkind, vulgar or offending messages, comments or videos") in question IC181 were recoded into the index of "distress from online content and cyberbullying". Each of the five items included five response options ("this did not happen to me", "not at all upset", "a little upset", "quite upset", "very upset").

Responses were assigned a value of 1 when students chose "not at all upset", a value of 2 when they chose "a little upset", a value of 3 when they chose "quite upset" and a value of 4 when they chose "very

upset". Responses of "this did not happen to me" were recoded to missing (9). Values across all items were then summed.

The index ranges from 0 to 20, with higher values indicating student experiencing more distress from online content and cyberbullying.

Items 1 to 4 were also used in the PISA 2022 ICT Questionnaire, while item 5 was added in PISA 2025.

Derived variables based on IRT scaling

The ICT Familiarity Questionnaire provided data for 12 DVs based on IRT scaling. The Cronbach's alpha for each scale and group are presented in Annex Table 22.A.45, and the number of items with international parameters for each scale and group are presented in Annex Table 22.A.46.

ICT availability at school (ICTSCH)

Students' frequency ratings of how often they use various digital resources at school (e.g., "desktop or laptop computer", "smartphone (i.e., mobile phone with internet access)") in question IC170 were scaled into the index of "ICT availability at school". Note that this scale is linked to the scale ICTSCH in PISA 2022, with parameters anchored for the seven link items. Each of the 10 items included in this scale had six response options ("never or almost never", "about once or twice a month", "about once or twice a week", "every day or almost every day", "several times a day", "this resource is not available to me outside of school"). "This resource is not available to me at school" was recoded as 0, while the five other response options were recoded as 1 prior to scaling. Annex Table 22.A.47 shows the item wording and item parameters for the items in this scale.

ICT availability outside school (ICTHOME)

Students' frequency ratings of how often they use various digital resources outside of school (e.g., "desktop or laptop computer", "smartphone (i.e., mobile phone with internet access)") in question IC171 were scaled into the index of "ICT availability outside school". Note that this scale is linked to the scale ICTHOME in PISA 2022, with parameters anchored for six link items. Each of the ten items included in this scale had six response options ("never or almost never", "about once or twice a month", "about once or twice a week", "every day or almost every day", "several times a day", "this resource is not available to me outside of school"). "This resource is not available to me outside of school" was recoded as 0, while the five other response options were recoded as 1 prior to scaling. Annex Table 22.A.48 shows the item wording and item parameters for the items in this scale.

Quality of access to ICT (ICTQUAL)

Students' ratings of their agreement with various statements about ICT resources at their school (e.g., "there are enough digital devices with access to the Internet at my school.", "the school's Internet speed is sufficient.") in question IC172 were scaled into the index of "quality of access to ICT". Note that this scale is linked to the scale ICTQUAL in PISA 2022, with parameters anchored for six link items. Each of the seven items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.49 shows the item wording and item parameters for the items in this scale.

Support or feedback via ICT (ICTFEED)

Students' frequency ratings of how often they use digital resources in various activities related to support or feedback (e.g., "read or listen to feedback sent by my teachers regarding my work and academic results", "read or listen to feedback sent by other students on my work") in question IC175 were scaled into

the index of “Support or feedback via ICT”. Note that this scale is linked to the scale ICTFEED in PISA 2022, with parameters anchored for four link items. Each of the four items included in this scale had five response options (“never or almost never”, “about once or twice a year”, “about once or twice a month”, “about once or twice a week”, “every day or almost every day”). Annex Table 22.A.50 shows the item wording and item parameters for the items in this scale.

Use of ICT for school activities outside of the classroom (ICTOUT)

Students’ frequency ratings of how often they use digital resources for various school-related activities outside of the classroom (e.g., “see my grades or results from specific assignments (e.g., homework or tests)”, “communicate with my teacher”) in question IC176 were scaled into the index of “Use of ICT for school activities outside of the classroom”. Note that this scale is linked to the scale ICTOUT in PISA 2022, with parameters anchored for eight link items. Each of the eight items included in this scale had five response options (“Never or almost never”, “About once or twice a year”, “About once or twice a month”, “About once or twice a week”, “Every day or almost every day”). Annex Table 22.A.51 shows the item wording and item parameters for the items in this scale.

Frequency of ICT activity – Weekday (ICTWKDY)

Students’ frequency ratings of how often they did various leisure activities using ICT during a typical week day (e.g., “play video-games (using my smartphone, a gaming console or an online platform or apps)”, “look for practical information online (e.g., find a place, book a train ticket, buy a product)”) in question IC177 were scaled into the index of “Frequency of ICT activity – Weekday”. Note that this scale is linked to the scale ICTWKDY in PISA 2022, with parameters anchored for seven link items. Each of the seven items included in this scale had six response options (“No time at all”, “Less than 1 hour a day”, “Between 1 and 3 hours a day”, “More than 3 hours and up to 5 hours a day”, “More than 5 hours and up to 7 hours a day”, “More than 7 hours a day”). Annex Table 22.A.52 shows the item wording and item parameters for the items in this scale.

Frequency of ICT activity – Weekend (ICTWKEND)

Students’ frequency ratings of how often they did various leisure activities using ICT during a typical weekend day (e.g., “play video-games (using my smartphone, a gaming console or an online platform or apps)”, “look for practical information online (e.g., find a place, book a train ticket, buy a product)”) in question IC178 were scaled into the index of “Frequency of ICT activity – Weekend”. Note that this scale is linked to the scale ICTWKEND in PISA 2022, with parameters anchored for seven link items. Each of the seven items included in this scale had six response options (“No time at all”, “Less than 1 hour a day”, “Between 1 and 3 hours a day”, “More than 3 hours and up to 5 hours a day”, “More than 5 hours and up to 7 hours a day”, “More than 7 hours a day”). Annex Table 22.A.53 shows the item wording and item parameters for the items in this scale.

Views of regulated ICT use in school (ICTREG)

Students’ ratings of their agreement with various statements about regulation of ICT use at school (e.g., “students should not be allowed to bring mobile phones to class”, “teachers should monitor what students do on their laptop or tablet device”) in question IC179 were scaled into the index of “views of regulated ICT use in school”. Note that this scale is linked to the scale ICTREG in PISA 2022, with parameters anchored for five link items. Each of the six items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.54 shows the item wording and item parameters for the items in this scale.

Students' practices regarding online information (ICTINFO)

Students' ratings of their agreement with various statements about their practices regarding online information (e.g., "when searching for information online I compare different sources", "I discuss the accuracy of online information with friends or other students") in question IC180 were scaled into the index of "students' practices regarding online information". Note that this scale is linked to the scale ICTINFO in PISA 2022, with parameters anchored for six link items. Each of the six items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.55 shows the item wording and item parameters for the items in this scale.

Interest in learning about digital resources (DIGINTLRN)

Students' ratings of their agreement with various statements about their interest in learning about digital resources (e.g., "I am interested in learning more about <digital resources>", "I like using <digital resources> to find new ways to do things") in question IC182 were scaled into the new PISA 2025 index of "interest in learning about digital resources". Each of the five items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.56 shows the item wording and item parameters for the items in this scale.

Interest in ICT (INTICT)

Students' ratings of their agreement with various statements about their interest and enjoyment of using ICT resources (e.g., "I forget about time when I'm using digital devices", "the Internet is a great resource for obtaining information I am interested in (e.g., news, sports, dictionary)") in question IC400 were scaled into the index of "interest in ICT". Note that this scale is linked to the scale INTICT in PISA 2018, with parameters anchored for six link items. Each of the six items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.57 shows the item wording and item parameters for the items in this scale.

ICT as a topic of social interaction (SOIAICT)

Students' ratings of their agreement with various statements about their experience with digital resources (e.g., "to learn something new about digital devices, I like to talk about them with my friends", "I like to meet friends and play computer and video games with them") in question IC401 were scaled into the index of "interest in ICT". Note that this scale is linked to the scale INTICT in PISA 2018, with parameters anchored for six link items. Each of the six items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.58 shows the item wording and item parameters for the items in this scale.

Parent Questionnaire indices

The Parent Questionnaire is an international option that countries/economies could choose to implement. It was administered to the parents of students participating in the PISA assessment. There were 12 variables derived from this questionnaire, including one simple DV and 11 IRT-scaled DVs. An overview of all DVs in this questionnaire is shown in Annex Table 22.A.59 and each of them is described in the following two sections.

Simple questionnaire indices

Parents' expectations in child's future educational career (PAREXPT)

Parents' responses to a list of possible educational levels they expected their children to complete in question PA183 were transformed into the index of "Parents' expectations in child's future educational career". The categories were specified using country-specific terms that were understood by the respondents. Each qualification was mapped to the ISCED classification of educational levels [see *ISCED 2011 Operational Manual: Guidelines for Classifying National Educational Programmes and Related Qualifications* (OECD/Eurostat/UNESCO Institute for Statistics, 2015)]. Values on the index ranged from "Less than ISCED level 2" to "ISCED level 8" and scores were assigned as noted in Annex Table 22.A.60.

Derived variables based on IRT scaling

The Parent Questionnaire provided data for 13 DVs based on IRT scaling. The Cronbach's alpha for each scale and group are presented in Annex Table 22.A.61, the number of items with international parameters for each scale and group are presented in Annex Table 22.A.62, and the number of trend items with international parameters for each trend scale and group are presented in Annex Table 22.A.63.

Child's past science activities (PRESUPP)

Parents' frequency ratings of how often their child at age 10 engaged in science-related (e.g., "watched TV programmes about science," "fixed broken objects or items, e.g. broken electronic toys") in question PA002 were scaled into the index of "child's past science activities". Note that this scale was linked to the scale PRESUPP in PISA 2015 through anchoring parameters for 10 link items. Each of the 13 items included in this scale had four response options ("very often", "regularly", "sometimes", "never"). All items were reverse-coded prior to scaling. Annex Table 22.A.64 shows the item wording and item parameters for the items in this scale and also indicates which items are trend items.

Family support for general learning (FGENSUPP)

Parents' frequency ratings of how often they or someone else in their home provides education-related support (e.g., "discuss how well my child is doing at school," "talk to my child about any problems he/she may have at school") in question PA003 were scaled into the new PISA 2025 index of "Current parental/guardian support". Note that while the item set used to derive this scale included seven items from PISA 2022 and PISA 2018, due to the still relatively low common items (seven out of 14) and the inclusion of domain-specific content in the previous cycle, it is not linked to the scale CURSUPP from PISA 2022 and 2016. Each of the seven items included in this scale had five response options ("never or hardly ever", "once or twice a year", "once or twice a month", "once or twice a week", "every day or almost every day"). Annex Table 22.A.65 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items. This scale was also derived from the FLA parent questionnaire using anchored parameters and set to the same metric.

Family support for science learning (FSCISUP)

Parents' or guardians' frequency ratings of how often they or someone else in their home provide education-related support (e.g., "help my child with his/her science homework", "discuss with my child how science can be applied in everyday life") in question PA003 were scaled into the new PISA 2025 index of "family support for science learning" (FSCISUP). Note that while the item set used to derive this scale included three items from PISA 2015, due to the relatively low number of common items (three out of eight) it is not linked to the scale CURSUPP in that cycle. Each of the four items included in this scale had five response options ("never or hardly ever", "once or twice a year", "once or twice a month", "once or twice a

week”, “every day or almost every day”). Annex Table 22.A.66 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA parent questionnaire using anchored parameters and set to the same metric.

Parental involvement (PARINVOL)

Parents’ responses to questions about their involvement in their child’s schooling in the past year (e.g., “Discussed my child’s progress with a teacher on my own initiative”, “Attended a scheduled meeting or conferences for parents”) in question PA008 were scaled into the index of “Parental involvement”. Note that this scale was linked to the PARINVOL scale in PISA 2022 through anchoring the parameters for nine link items. Each of the 11 items included in this scale had two substantive response options (“Yes”, “No”) and an additional response option “Not supported by school” which was recoded as missing prior to scaling. All items were reverse-coded so that higher scale scores indicate higher levels of involvement. Annex Table 22.A.67 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA parent questionnaire using anchored parameters and set to the same metric.

Parental views of digital resource use at school (PVDRSCUSE)

Parents’ ratings of their agreement with statements about the use of digital resources at school (e.g., “parents should use <digital resources> to communicate with their child’s teachers and school”, “schools should allow parents to track the progress of their children at school using <digital resources>”) in question PA203 were scaled into the new PISA 2025 index of “parental views of digital resource use at school”. Each of the five items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.68 shows the item wording and item parameters for the items in this scale.

Parents’ concerns regarding environmental topics (PQENPERC)

Parents’ ratings of their level of concern about environmental issues (e.g., air pollution”, “clearing of forests for other land use”) in question PA035 were scaled into the index of “parents’ concerns regarding environmental topics”. Note that this scale was linked to the PQENPERC scale in PISA 2015 through anchoring the parameters for seven link items. Each of the 11 items included in this scale had four response options (“this is a serious concern for me personally as well as others”, “this is a serious concern for other people in my country but not for me personally”, “this is a serious concern only for people in other countries”, “this is not a serious concern for anyone”). All items were reverse-coded prior to scaling so that so that higher scale scores correspond to higher levels of parents’ concerns regarding environmental topics. Annex Table 22.A.69 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Parents’ perceptions of students’ self-regulated learning skills (PQSELFREG)

Parents’ ratings of their agreement with statements about their child’s self-regulated learning skills (e.g., “my child is curious to learn new things on his/her school subjects”, “my child asks questions whenever he/she does not understand something”) in question PA204 were scaled into the index of “parents’ perceptions of students’ self-regulated learning skills”. Each of the seven items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.70 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA parent questionnaire using anchored parameters and set to the same metric.

Parents' views on broad science (PQGENSCI)

Parents' ratings of their agreement with statements about the importance of broad science (e.g., "<broad science> is important to help us to understand the natural world", "I find that <broad science> helps me to understand the things around me") in question PA033 were scaled into the index of "parents' views on broad science". Note that this scale was linked to the PQVSCIE scale in PISA 2015 through the anchoring parameters for five link items. Each of the five items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). All items were reverse-coded prior to scaling so that higher scale scores reflect higher levels of agreement. Annex Table 22.A.71 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Parents' views on future environmental topics (PQENVOPT)

Parents' ratings of their expectations about the future of environmental issues (e.g., air pollution, "clearing of forests for other land use") in question PA036 were scaled into the index of "parents' view on future environmental topics". Note that this scale was linked to the PQENVOPT scale in PISA 2015 through anchoring the parameters for five link items. Each of the five items included in this scale had three response options ("improve", "stay about the same", "get worse"). All items were reverse coded prior to scaling so that higher scale scores indicate higher levels of parent's environmental optimism. Annex Table 22.A.72 shows the item wording and item parameters for the items in this scale.

Parents' digital self-efficacy (PQDIGEFF)

Parents' ratings of their agreement with statements regarding their confidence in doing tasks with digital resources (e.g., "create a multi-media presentation (with sound, pictures, or video)", "build or edit a webpage (e.g. using <WordPress>") in question PA205 were scaled into the new PISA 2025 index of "parents' digital self-efficacy". Each of the nine items included in this scale had four response options ("I cannot do this", "I struggle to do this on my own", "I can do with a bit of effort", "I can easily do this", "I don't know what this is"). The last category was coded as 0, while the other categories were coded sequentially from 1 ("I cannot do this") to 4 ("I can easily do this"). Annex Table 22.A.73 shows the item wording and item parameters for the items in this scale.

Positive parental feedback (PQFEED)

Parents' ratings of their agreement with statements about parental feedback (e.g., "when my child does well I tell them they worked hard", "If my child did well, I would be proud of how hard they tried") in question PA401 were scaled into the new PISA 2025 index of "positive parental involvement". Each of the five items included in this scale had four response options ("strongly disagree", "disagree", "agree", "strongly agree"). Annex Table 22.A.74 shows the item wording and item parameters for the items in this scale.

School quality (PQSCHOOL)

Parents' ratings of their agreement with statements about school quality (e.g., "most of my child's school teachers seem competent and dedicated.", "standards of achievement are high in my child's school.") in question PA007 were scaled into the index of "school quality". Note that through anchoring of parameters for six items this scale was linked to the PQSCHOOL scale in PISA 2022, which was in turn linked to PISA 2018. Each of the eight items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). All items were reverse-coded so that higher scale scores reflect higher levels of agreement. Annex Table 22.A.75 shows the item wording and item parameters for the items in this scale.

Science-related career (PQSCAR)

Parents' responses to questions about science-related careers (e.g., "does anybody in your family (including you) work in a <science-related career>?", "does your child show an interest in working in a <science-related career>?") in question PA032 were scaled into the index of "science-related career". Each of the five items included in this scale had two response options ("Yes", "No"). Annex Table 22.A.76 shows the item wording and item parameters for the items in this scale.

School Questionnaire indices

The School Questionnaire consisted mainly of questions used in previous cycles. There were 45 variables derived from this questionnaire, including 28 simple DVs and 17 IRT-scaled DVs. An overview of all DVs in this questionnaire is shown in Annex Table 22.A.77 and each are described in the following sections. The simple and scaled DVs are organised first by framework module (please see Chapter 6 for a description of modules) and then alphabetical order within modules.

Simple questionnaire indices

School type and resources (Module 1)

Percentage of students enrolled in the school (SC002Q01TA_P and SC002Q02TA_P)

School principals were asked in SC002 to report the number of students enrolled in their school, separately for boys and for girls. The SC002Q01TA_P and SC002Q02TA_P indices are computed as the percentage of boys and girls (respectively) enrolled in the school using the sum of the two values as the total.

Migration and language exposure (Module 4) and School culture and climate (Module 6)

Percentage of students with specific characteristics (SC211Q02JA_Q to SC211Q06JA_Q)

School principals were asked in SC211 to report the percentage of students with special needs (Q02), from socioeconomically disadvantaged homes (Q03) and who are refugees (Q06). The international distribution of responses to the items in SC211 was divided into four equal groups (quartiles). Indices SC211Q02JA_Q to SC211Q06JA_Q contain principal's responses classified into one of those 4 groups.

Out-of-school experiences (Module 10)

Creative extra-curricular activities (CREACTIV)

School principals were asked in SC053 to report what extra-curricular activities their schools offered to 15-year-old students. The two response categories were "Yes" and "No" for the 10 items. The index of creative extra-curricular activities at school (CREACTIV) was computed as the total number of the following 3 activities that occurred at school: i) band, orchestra or choir (SC053Q01TA); ii) school play or school musical (SC053Q02TA); and iii) art club or art activities (SC053Q09TA). The index ranges from 0 to 3.

School type and resources (Module 11)

Availability of computers (RATCMP1)

School principals were asked in SC004 to report the number of digital devices available for 15-year-old students at their school. The index of availability of computers (RATCMP1) is the ratio of the number of

desktop or laptop computers available for these students for educational purposes (SC004Q02TA) to the total number of students in the modal grade for 15-year-olds at their school (SC004Q01TA).

Availability of tablet and e-book readers devices (RATTAB)

School principals were asked in SC004 to report the number of tablet devices and e-book readers available for 15-year-old students at their school for educational purposes (SC004Q08DA and SC004Q09DA). The index of availability of tablet devices (RATTAB) is the ratio of the number of tablet and e-book readers devices available for these students for educational purposes to the total number of students in the modal grade for 15-year-olds at their school (SC004Q01TA).

Computers connected to the Internet (RATCMP2)

School principals were asked in SC004 to report the number of desktop or laptop computers at their school that are connected to the Internet. The index of computers connected to the Internet (RATCMP2) is the ratio of the number of desktop or laptop computers available for 15-year-olds for educational purposes (SC004Q02TA) to the number of these computers that are connected to the Internet (SC004Q03TA).

Proportion of female science teachers at school (PROFEMST)

The proportion of female science teachers was computed as the total number of full-time and part-time female science teachers at their school provided by the principal in SC400 (SC400Q02DA_01, SC400Q02DA_02), divided by the total number of science teachers at their school provided by the principal in SC400 (SC400Q01DA_01, SC400Q02DA_02). These proportions were recoded into the PROFEMST index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of fully certified science teachers at school (PROSTCE)

The proportion of fully certified science teachers was computed as the total number of full-time and part-time fully certified science teachers at their school provided by the principal in SC400 (SC400Q03DA_01, SC400Q03DA_02), divided by the total number of science teachers at their school provided by the principal in SC400 (SC400Q01DA_01, SC400Q02DA_02). These proportions were recoded into the PROSTCE index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of personnel for pedagogical support (PROPSUPP)

Principals were asked in SC168 to report the number of personnel for pedagogical and well-being support currently working in their school (SC168Q01DA and SC168Q05DA), respectively. The proportion of personnel for pedagogical support (PROPSUPP) was calculated by dividing the number of these personnel by the total number of non-teaching staff at the school (TOTSTAFF). These proportions were recoded into the PROSTCE index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of school administrative personnel (PROADMIN)

Principals were asked in SC168 to report the number of school administrative personnel currently working in their school (SC168Q02JA). The proportion of school administrative personnel was calculated by dividing the number of these personnel by the total number of non-teaching staff at the school (sum of SC168Q01DA, SC168Q05DA, SC168Q02DA and SC168Q03JA). These proportions were recoded into the PROADMIN index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of school management personnel (PROMGMT)

Principals were asked in SC168 to report the number of school management personnel currently working in their school (SC168Q03JA). The proportion of school management personnel was calculated by dividing the number of these personnel by the total number of non-teaching staff at the school (sum of SC168Q01DA, SC168Q05DA, SC168Q02DA and SC168Q03JA). These proportions were recoded into the PROMGMT index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of science teachers at school (PROPSCIE)

The proportion of science teachers was computed as the total number of full-time and part-time science teachers at their school provided by the principal in SC400 (SC400Q01DA_01, SC400Q01DA_02), divided by the total number of teachers at their school provided by the principal in SC018 (SC018Q01TA_01, SC018Q01TA_02). These proportions were recoded into the PROPSCIE index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of staff focused on well-being (PROWBST)

Principals were asked in SC168 to report the number of school personnel focused on student well-being (eg, psychologists, nurses, and counsellors) currently working in their school (SC168Q05DA). The proportion of staff focused on well-being was calculated by dividing the number of these personnel by the total number of non-teaching staff at the school (sum of SC168Q01DA, SC168Q05DA, SC168Q02DA and SC168Q03JA). These proportions were recoded into the PROWBST index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

School size (SCHSIZE)

School size is calculated from the enrolment data provided by the school principal in SC002, summing the number of girls (SC002Q02TA) and boys (SC002Q01TA) at a school. If countries included a national addition in SC002, the number of students in that category was also considered for calculating school size.

The international distribution of school size was divided into four equal groups (quartiles). The SCHSIZE index contains principal's responses classified into one of those 4 groups.

School type (SCHLTYPE)

Schools are classified as either public or private according to whether a private entity or a public agency has the ultimate power for decision making concerning its affairs. As in previous PISA surveys, the index of school type (SCHLTYPE) was constructed by recoding SC013 and SC016. SC013 asks whether the school is public or private, and SC016 asks about the source and proportion of resources (government, resources from students/parents, benefactors/donations, or other). SCHLTYPE has the following three categories:

1. Private independent (if SC013Q01TA=2 and SC016Q01TA < 50), or (SC013Q01TA=2 and SUM(SC016Q02TA, SC016Q03TA, SC016Q04TA) >= 50).
2. Private Government-dependent (if SC013Q01TA=2 and SC016Q01TA >= 50), or
3. Public (if SC013Q01TA=1).

Since PISA 2018, PRIVATESCH was created from sampling information in order to improve the public/private indicators. If SC013 is missing, PRIVATESCH is used to create SCHLTYPE.

Similar to 2018, IRL had special treatment for this designation – based solely on the STRATUM sampling variable.

Student-teacher ratio (STRATIO)

The student-teacher ratio was obtained by dividing the number of enrolled students provided by the principal in SC002 by the total number of full-time and part-time teachers provided by the principal in SC018 (SC018Q01TA_01, SC018Q01TA_02). The international distribution of student-teacher ratio was divided into four equal groups (quartiles). The STRATIO index contains the student-teacher ratio classified into one of those 4 groups.

Student-science teacher ratio (SCRATIO)

The student-science teacher ratio (SCRATIO) was obtained by dividing the number of students (SCHSIZE) provided by the principal in SC002 by the total number of full-time and part-time science teachers provided by the principal in SC400 (SC400Q01DA_01, SC400Q01DA_02).

Total number of science teachers at school (TOTSCIE)

Principals were asked in SC400 to report the number of full-time and part-time mathematics teachers at their school (SC400Q01DA_01, SC400Q01DA_02) and provide additional information on how many of the staff was full-time and part-time employed mathematics teachers qualified at different ISCED levels. The total number of science teachers at the school was computed as the sum of full-time and part-time science teachers. The international distribution of the total number of science teachers was divided into four equal groups (quartiles). The TOTSCIE index contains principal's responses classified into one of those 4 groups.

Total number of non-teaching staff at school (TOTSTAFF)

Principals were asked in SC168 to report the number of non-teaching staff currently working in their school. The total number of non-teaching staff at the school was calculated as the sum of the number of personnel for pedagogical and well-being support (SC168Q01DA and SC168Q05DA), school administrative personnel (SC168Q02DA) and school management personnel (SC168Q03JA). The international distribution of the total number of non-teaching staff was divided into four equal groups (quartiles). The TOTSTAFF index contains principal's responses classified into one of those 4 groups.

Total number of all teachers at school (TOTAT)

Principals were asked in SC018 to report the total number of full-time and part-time teachers at their school (SC018Q01TA_01, SC018Q01TA_02) and provide additional information on how many of the staff was full-time and part-time employed teachers qualified at different ISCED levels.

Proportion of funding coming from different sources (SC016Q01TA_Q to SC016Q04TA_Q)

Principals were asked in SC016 to report the percentage of funding their schools receive from different sources. Responses for each item were recoded into four categories: 0% to 25%; 26% to 50%; 51% to 75%; More than 75%. Indices SC016Q01TA_Q to SC016Q04TA_Q contain principal's responses classified into one of those 4 groups.

Availability of digital devices (SC004Q01TA_Q to SC004Q09DA_Q)

Principals were asked in SC004 to report the number of students in the modal grade for 15 year-old students and the number of a variety of digital devices in their schools. This information is primarily used

to produce the RATCOMP1, RATCOMP2 and RATAB indices described above. The international distribution of responses to the items in SC004 was divided into four equal groups (quartiles). Indices SC004Q01TA_Q to SC004Q09DA_Q contain principal's responses classified into one of those 4 groups.

Selection and enrolment (Module 12)

School selectivity (SCHSEL)

Principals were asked in SC012 about admittance policies at their school, including student academic performance and recommendation by feeder schools. The three response categories for this question were “Never”, “Sometimes”, and “Always”. An index of academic school selectivity (SCHSEL) was computed by assigning schools to one of three categories based on how often two factors, namely “Student’s record of academic performance” (SC012Q01TA) and “Recommendation of feeder schools” (SC012Q02TA), were considered when admitting students to the school as follows:

1. the two factors (student’s record of academic performance and recommendation of feeder schools) were never considered (if SC012Q01TA=1 and SC012Q02TA=1),
2. at least one of the factors was considered sometimes but neither always (if SC012Q01TA=2 or SC012Q02TA=2, and if SC012Q01TA<3 and SC012Q02TA<3), and
3. at least one of the factors was considered always (if SC012Q01TA=3 or SC012Q02TA=3).

Organisation of student learning at school (Module 14)

Ability grouping for science classes (ABGSCIE)

School principals were asked in SC404 to report the extent to which their science classes catered to students with different abilities. The three response categories were “For all classes”, “For some classes”, and “Not for any classes”. An index of ability grouping between science classes (ABGSCIE) was derived from the first two items (SC404Q01DA, SC404Q02DA) by assigning schools to three categories: (1) schools with no ability grouping for any classes, (2) schools with one of these forms of ability grouping between some classes and (3) schools with one of these forms of ability grouping for all classes.

Class size (CLSIZE)

Principals were asked in SC003 about the average size of the test language classes in their school. The nine response categories were “15 students or fewer”, “16-20 students”, “21-25 students”, “26-30 students”, “31-35 students”, “36-40 students”, “41-45 students”, “46-50 students”, and “More than 50 students”. The average class size (CLSIZE) was derived from the midpoint of each response category, resulting in a value of 13 for the lowest category, and a value of 53 for the highest.

Science class size (SCLSIZE)

Principals were asked in SC403 about the average class size of science classes in their school. The nine response categories were “15 students or fewer”, “16-20 students”, “21-25 students”, “26-30 students”, “31-35 students”, “36-40 students”, “41-45 students”, “46-50 students”, and “More than 50 students”. The average science class size was derived from the midpoint of each response category, resulting in a value of 13 for the lowest category, and a value of 53 for the highest.

Teacher qualification, training, and professional development (Module 17)

Proportion of all teachers fully certified (PROATCE)

Principals were asked in SC018 to report the number of full-time and part-time teachers fully certified by the appropriate authority (SC018Q02TA_01, SC018Q02TA_02). The proportion of fully certified teachers was computed by dividing the number of fully certified teachers by the total number of teachers. These proportions were recoded into the PROATCE index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of all teachers with at least ISCED level 6 bachelor qualification (PROPAT6)

Principals were asked in SC018 to report the number of full-time and part-time teachers with an ISCED level 6 (Bachelor's or equivalent level) qualification (SC018Q08JA_01, SC018Q08JA_02). The proportion of teachers with *at least* an ISCED 6 Bachelor qualification was calculated by dividing the number of full-time and part-time teachers with an ISCED level 6 (Bachelor's or equivalent level) qualification (SC018Q08JA_01, SC018Q08JA_02), ISCED level 7 (Master's or equivalent level) qualification (SC018Q09JA_01, SC018Q09JA_02), and ISCED level 8 (Doctoral or equivalent level) qualification (SC018Q10JA_01, SC018Q10JA_02) by the total number of teachers (sum of SC018Q01TA_01 and SC018Q01TA_02). These proportions were recoded into the PROPAT6 index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of all teachers with at least ISCED level 7 master qualification (PROPAT7)

Principals were asked in SC018 to report the number of full-time and part-time teachers with an ISCED level 7 (Master's or equivalent level) qualification (SC018Q09JA_01, SC018Q09JA_02). The proportion of teachers with *at least* an ISCED 7 Master qualification was calculated by dividing the number of full-time and part-time teachers with an ISCED level 7 (Master's or equivalent level) qualification (SC018Q09JA_01, SC018Q09JA_02) and ISCED level 8 (Doctoral or equivalent level) qualification (SC018Q10JA_01, SC018Q10JA_02) by the total number of teachers (sum of SC018Q01TA_01 and SC018Q01TA_02). These proportions were recoded into the PROPAT7 index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Proportion of all teachers with ISCED level 8 doctoral qualification (PROPAT8)

Principals were asked in SC018 to report the number of full-time and part-time teachers with an ISCED level 8 (Doctoral or equivalent level) qualification (SC018Q10JA_01, SC018Q10JA_02). The proportion of teachers with an ISCED 8 Doctoral qualification (PROPAT8) was calculated by dividing the number of these teachers by the total number of teachers (sum of SC018Q01TA_01 and SC018Q01TA_02). These proportions were recoded into the PROPAT8 index, which has five categories: 0 to 0.2; higher than 0.2 and up to 0.4; higher than 0.4 and up to 0.6; higher than 0.6 and up to 0.8; higher than 0.8 and up to 1.

Parental involvement and support (Module 19)

Proportion of parents participating in school-related activities (SC064Q01TA_Q to SC064Q07WA_Q)

Principals were asked in SC064 to report the proportion of students' parents or guardians that participated in specific school-related activities in the last academic year. The international distribution of responses in

each item was divided into four equal groups (quartiles). Indices SC064Q01TA_Q to SC064Q07WA_Q contain principal's responses classified into one of those 4 groups.

Derived variables based on IRT scaling

The School Questionnaire provided data for 25 DVs based on IRT scaling. The Cronbach's alpha for each scale and group are presented in Annex Table 22.A.78, the number of items with international parameters for each scale and group are presented in Annex Table 22.A.79, the number of trend items with international parameters for each trend scale and group are presented in Annex Table 22.A.80, the countries/economies for which the scale scores were suppressed for each scale are presented in Annex Table 22.A.81, and the groups that did not administer each scale are presented in Annex Table 22.A.82.

PISA preparation and effort (Module 5)

School support for PISA administration (PISASUPP)

Principals were asked in SC209 whether the school principal had supported the PISA administration through different activities (e.g., "I have shared information about the PISA test administration with the staff at my school", "I have spoken to teachers at my school about the importance of the PISA test"). The six items had two response categories ("yes", "no") and were used to derive the new PISA 2025 index "school support for PISA administration (PISASUPP)". Higher scale score values indicate higher levels of support for the administration of PISA at school. Annex Table 22.A.83 shows the item wording and item parameters for the items in this scale.

School culture and climate (Module 6)

Negative school climate (NEGSCCLIM)

Principals were asked in SC172 about the extent of problem behaviours that contribute to a negative school climate in their school (e.g., "Profanity", "Vandalism"). The four response categories for the eight items in the scale were "Not at all", "Small extent", "Moderate extent", and "Large extent" and were used to derive the index "negative school climate", which is linked to the PISA 2022 scale NEGSCCLIM through anchoring parameters for six link items. Higher scale score values indicate that problem behaviours contribute to a negative school climate to a greater extent, while lower scale score values indicate that problem behaviours impact school climate to a lesser extent. Annex Table 22.A.84 shows the item wording and item parameters for the items in this scale.

Shortage of educational material (EDUSHORT)

Principals were asked in SC017 about the extent to which instruction is hindered by a shortage of educational materials in their school (e.g., "A lack of educational material (e.g., textbooks, IT equipment, library or laboratory material)", "Inadequate or poor quality educational material (e.g. textbooks, IT equipment, library or laboratory material)"). Note that this scale was linked to the EDUSHORT scale in PISA 2018. The four response categories for the four items in the scale were "Not at all", "Very little", "To some extent", and "A lot". Higher scale score values indicate that the school is impacted by a shortage of educational materials to a greater extent, while lower scale score values indicate that the school is impacted to a lesser extent by a shortage of educational materials. Annex Table 22.A.85 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Shortage of educational staff (STAFFSHORT)

Principals were asked in SC017 about the extent to which instruction is hindered by a shortage of educational staff in their school (e.g., “A lack of teaching staff”, “Inadequate or poorly qualified assisting staff”). Note that this scale was linked to the STAFFSHORT scale in PISA 2018. The four response categories for the four items in the scale were “Not at all”, “Very little”, “To some extent”, and “A lot”. Higher scale score values indicate that the school is impacted by a shortage of educational staff to a greater extent, while lower scale score values indicate that the school is impacted to a lesser extent by a shortage of educational staff. Annex Table 22.A.86 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Science identity (Module 7)

Opportunities for environment-related activities at school (OPPACTENV)

Principals were asked in SC250 whether the school offered students the opportunity to take part environment-related activities (e.g., “whole school or community rubbish collection”, “showcasing environmental achievement and policies in school newsletters, on websites and school displays”). The 14 items had four response categories (“never”, “sometimes”, “frequently”, and “very frequently”) and were used to derive the new PISA 2025 index “opportunities for environment-related activities at school (OPPACTENV)”. Higher scale score values indicate higher levels of opportunities for students at the school. Annex Table 22.A.87 shows the item wording and item parameters for the items in this scale.

Out-of-school experiences (Module 10)

Extra-curricular activities at school (ALLACTIV)

School principals were asked in SC053 to report what extra-curricular activities their schools offered to 15-year-old students (e.g., “School play or school musical”, “Mathematics club”). The 11 items with two response categories (“Yes” and “No”) were used to derive the index “extra-curricular activities at school”, which is linked to the PISA 2022 scale ALLACTIV through anchoring parameters for the eight link items. Higher scale score values indicate that more extra-curricular activities were offered by the school, while lower scale score values indicate that fewer extra-curricular activities were offered. Annex Table 22.A.88 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

School type and infrastructure (Module 11)

Availability of learning software (AVRLSOFT)

School principals were asked in SC261 to report the percentages of digital devices the school had access to (e.g., “software for creating a multi-media presentation (e.g. <Microsoft® PowerPoint™>”, “software for programming (e.g. <Scratch>, <Python>, <Java>)”). The 10 items with five response categories (“none”, “1-25%”, “26-50%”, “51-75%”, and “76-100%”) were used to derive the index “Availability of learning software” (AVRLSOFT), which is new to PISA 2025. Higher scale score values indicate that higher levels of availability of software at school. Annex Table 22.A.89 shows the item wording and item parameters for the items in this scale.

Preparedness for digital learning (DIGPREP)

School principals were asked in SC155 to indicate their agreement with statements about the school’s preparedness for digital learning (e.g., “the number of digital devices connected to the Internet is sufficient”,

“teachers have the necessary technical and pedagogical skills to integrate digital devices in instruction”). The 12 items had four response categories (“strongly disagree”, “disagree”, “agree”, “strongly agree”) and were used to derive the index “preparedness for digital learning” (DIGPREP), which is new to PISA 2025. Higher scale score values indicate that higher levels of preparedness for digital learning. Annex Table 22.A.90 shows the item wording and item parameters for the items in this scale.

School autonomy (Module 13)

Educational leadership (EDULEAD)

Principals were asked in SC201 about how often they or other members of their school management team engaged in activities or behaviours related to educational leadership during the past 12 months (e.g., “providing feedback to teachers based on observations of instruction in the classroom”, “taking actions to ensure that teachers feel responsible for their students’ learning outcomes”), and the seven items were used to measure the index “educational leadership”, which is linked PISA 2022 scale EDULEAD through anchoring six link items. The five response categories for the seven items in the scale were “Never or almost never”, “About once or twice a year”, “About once or twice a month”, “About once or twice a week”, and “Every day or almost every day”. Higher scale score values indicate higher frequencies of engagement by the principal and school management team in educational leadership activities, while lower scale values indicate lower frequencies of engagement by the principal and school management team in educational leadership activities. Annex Table 22.A.91 shows the item wording and item parameters for the items in this scale.

Organisation of student learning at school (Module 14)

Delivery of global issues in curriculum (DELGLOB)

School principals were asked in SC158 to indicate whether there are formal curricula for global issues at school (e.g., “climate change and global warming”, “international conflicts”). The seven items in this question had each two response categories (“yes”, “no”) and were used to derive the new PISA 2025 index “delivery of global issues in curriculum” (DELGLOB). Higher scale score values indicate higher frequencies of formal curricula for these topics. Annex Table 22.A.92 shows the item wording and item parameters for the items in this scale.

Priorities for learning objectives (PRIORLRN)

School principals were asked in SC266 to indicate what priority is given teaching objectives at school (e.g., “to enhance students’ capacity to learn on their own”, “to teach students how to use the feedback they receive from teachers”). The seven items in this question had each four response categories (“no priority”, “low priority”, “medium priority” and “high priority”) and were used to derive the new PISA 2025 index “priorities for learning objectives” (PRIORLRN). Higher scale score values indicate higher priorities, while lower scale score values indicate that these objectives had lower priority. Annex Table 22.A.93 shows the item wording and item parameters for the items in this scale.

Exposure to science content (Module 15)

Freedom of choice for studying science (FREESCIE)

School principals were asked in SC406 to indicate to which extent students can choose different aspects of science learning (e.g., “students can choose the <school science> course(s) they study”, “students can choose the level of difficulty”). The four items in this question had each three response categories (“they cannot choose at all”, “they can choose to a certain extent” and “they can choose freely”) and were used

to derive the new PISA 2025 index “freedom of choice for studying science” (FREESCIE). Higher scale score values indicate higher levels of freedom of choice. Annex Table 22.A.94 shows the item wording and item parameters for the items in this scale.

Teacher Qualification, Training, and Professional Development (Module 17)

Science teacher training (SCTRAIN)

School principals were asked in SC408 to indicate whether the school offered science teachers professional development in different areas (e.g., “school science content”, “integrating digital resources into science content”). The six items in this question had each two response categories (“yes” and “no”) and were used to derive the new PISA 2025 index “science teacher training” (SCTRAIN). Higher scale score values indicate higher levels of professional development offers for science teachers at school. Annex Table 22.A.95 shows the item wording and item parameters for the items in this scale.

Assessment, evaluation, and accountability (Module 18)

Use of SES assessments (SESASSESS)

Principals were asked in SC197 whether results from assessment of students’ social and emotional skills were used in different ways (e.g., “results are tracked over time by an administrative authority”, “results are compared to the results of other schools”). The six items had two response categories (“yes”, “no”) and were used to derive the new PISA 2025 index “use of SES assessments” (SESASSESS). Higher scale score values indicate greater use of SES assessment results. Annex Table 22.A.96 shows the item wording and item parameters for the items in this scale.

Use of standardised tests (STDTEST)

Principals were asked in SC035 to indicate whether standardised tests were used for various purposes (e.g., “To guide students’ learning”, “To group students for instructional purposes”). Eleven items with two response categories (“yes” and “no”) were used to derive the index “use of standardised tests”, which is linked to the PISA 2022 scale STDTEST. Higher scale score values indicate that standardised tests are used for accountability purposes to a greater extent, while lower scale score values indicate that these tests are used to a lesser extent. Annex Table 22.A.97 shows the item wording and item parameters for the items in this scale.

Use of teacher-developed tests (TDTEST)

Principals were asked in SC035 to indicate whether teacher-developed tests were used for various purposes (e.g., “To guide students’ learning”, “To group students for instructional purposes”). Eleven items with two response categories (“yes” and “no”) were used to derive the index “use of teacher-developed tests”, which is linked to the PISA 2022 scale TDTEST. Higher scale score values indicate that teacher-developed tests are used for accountability purposes to a greater extent, while lower scale score values indicate that these tests are used to a lesser extent. Annex Table 22.A.98 shows the item wording and item parameters for the items in this scale.

Parental/guardian involvement and support (Module 19)

School encouragement of parent or guardian involvement (ENCOURPG)

Principals were asked in SC192 about how often their school staff engaged parents or guardians in various aspects of students’ educational environment during the last academic year (e.g., “invited parents or

guardians to volunteer for school activities”, “initiated communications with parents or guardians about their child’s progress”). The six items in this question had four response categories (“never or almost never”, “a few times a year”, “a few times a month”, and “once a week or more”) and were used to derive the index “school encouragement of parent or guardian involvement, which is linked to the PISA 2022 scale ENCOURPG. Higher scale score values indicate more frequent efforts by the school staff to engage parents or guardians in becoming involved at the school, while lower scale score values indicate less frequent engagement efforts. Annex Table 22.A.99 shows the item wording and item parameters for the items in this scale.

Science Teacher Questionnaire indices

The Teacher Questionnaire is an international option that countries/economies could choose to implement. Routing within the questionnaire was used to deliver specific questions to mathematics teachers and specific questions to all other, non-mathematics teachers. Other questions in the questionnaire were administered to all teachers. There were 79 variables derived from this questionnaire, including 56 simple DVs and 23 IRT scaled DVs. An overview of all DVs in this questionnaire is shown in Annex Table 22.A.100 and each are described in the following sections.

Simple questionnaire indices

Originally trained teachers –Broad (OTT2) definitions

The Teacher Questionnaire included one question about teachers’ initial education and professional development. TC015 asked about how the teacher qualification was received. Response options included “1, I attended a standard teacher education or training programme at an <educational institute which is eligible to educate or train teachers>.”, “2, I attended an in-service teacher education or training programme.”, “3, I attended a work-based teacher education or training programme.”, “4, I attended training in another pedagogical profession.”, or “5, Other”. This question was used to build the DV OTT2 (Originally trained teachers, broad definition). The broad definition implies that a teacher has finished any of the following three programmes: either a “standard teacher education or training programme at an <educational institute which is eligible to educate or train teachers>” (option 1 in TC015), an “in-service teacher education or training programme” (option 2) or a “work-based teacher education or training programme” (option 3 in TC015).

Trained to teach certain subjects (NTEACH1-13)

TC018 asked about the specific subjects that were included in the teacher’s education or training programme or other professional qualification and asked if the respondents taught these subjects to the national modal grade for 15-year-olds in the current school year. The DVs NTEACH1 to NTEACH 13 reflect whether the teacher was trained to teach a certain subject. A value of “1” indicate that teachers were trained to teach the subject in question, while “0” indicates they were not trained to teach the subject in question.

Subject-specific overlap between initial education and teaching the modal grade (STTMG1-13)

TC018 enquired about the specific subjects that were included in the teacher’s education or training programme or other professional qualification and asked if the respondents taught these subjects to the national modal grade for 15-year-olds in the current school year. This question is used to build the DVs STTMG1 to STTMG13, indicating the subject-specific overlap between initial education and teaching the

modal grade, i.e., whether a teacher currently teaches a certain subject combined with whether it was included in the teacher's initial training. A value of "0" indicates that teachers were neither trained nor teach the subject in question, "1" indicates they were trained to teach the subject in question but do not teach it, "2" indicates they were not trained to teach the subject in question but they do teach the subject, and "3" indicates they were trained to teach the subject in question and they teach the subject.

Content overlap between initial education and professional development (TC045Q01-TC045Q24)

TC045 asked about 19 content topics (e.g., "knowledge and understanding of my subject field(s)", "knowledge of the curriculum") that might have been included in the teachers' initial education and training and/or in professional development activities during the last 12 months. Teachers could select both if applicable. The DVs TC045Q01 to TC045Q24 reflect the content overlap between initial education and professional development.

Higher educational level attained (TCISCED)

Teachers' responses to TC210 regarding education were classified using ISCED 2011. An index on higher educational level attained was constructed by recoding educational qualifications into the following categories: Less than ISCED Level 3.3 (upper secondary with no direct access to tertiary education), ISCED Level 3.3 (upper secondary with no direct access to tertiary education), ISCED 3.4 (upper secondary with direct access to tertiary education), ISCED 4 (post-secondary non-tertiary), ISCED 5 (short-cycle tertiary), ISCED 6 (Bachelor's or equivalent level), ISCED 7 (Master's or equivalent level), and ISCED 8 (Doctoral or equivalent level). Scores are assigned as noted in Annex Table 22.A.101.

Employment status (EMPLSTAT4 and EMPLSTATd4)

TC211 asked about employment status in terms of the contract duration with three response options ("Permanent employment", "Fixed-term contract for a period of more than 1 school year", "Fixed-term contract for a period of 1 school year or less"). The corresponding DVs reflected the duration of employment, measured via TC211, a) on the original four-point scale (EMPLSTAT4) and b) dichotomous, distinguishing a permanent position from other contracts (EMPLSTATd4).

Teacher workload (TC216Q01JA_Q to TC216Q08JA_Q)

Teachers were asked to enter how many hours a week they spend on various teaching tasks (e.g., "Marking/correcting of student work", "General administrative work"). Teachers' responses to each of the eight fill-in items in question TC216 were divided into four equal groups (quartiles). The indices TC216Q01JA_Q to TC216Q08JA_Q contain the responses classified into one of those 4 groups.

Weekly teacher workload (TCWKLOAD)

Teachers were asked to enter how many hours a week they spend on various teaching tasks (e.g., "Marking/correcting of student work", "General administrative work"). Teachers' responses to each of the eight fill-in items in question TC216 were summed and the international distribution of responses was divided into four equal groups (quartiles). The TCWKLOAD_Q index contains the weekly teacher workload classified into one of those 4 groups.

Use of digital resources for science (ICTSCIETC)

Teachers' frequency ratings of how often they instruct their students to use digital resources for a range of science tasks in class or for homework (e.g., "Use <digital resources> to make students observe a natural

system or phenomenon”, “Use <digital resources> for simulations and modelling (e.g. <GeoGebra>, <NetLogo>,)” in question TC428 were aggregated into a simple index “Use of digital resources for science” as follows: A value of “1” is assigned if teachers select response options 4 (“About once or twice a week”) or 5 (“Every day or almost every day”) at least once across the 4 items in this question, “0” otherwise. The index captures whether teachers said that they use some digital resources for science lessons at least on a weekly basis or not.

Teaching experience (ST007Q01NA_Q and ST007Q02NA_Q)

In TC007 teachers were asked to how many years of teaching experience they had at the school having them take the questionnaire and how many years of teaching experience they had in total. The international distribution of responses for each item in SC007 was divided into four equal groups (quartiles). The ST007Q01NA_Q and ST007Q02NA_Q indices contain principal’s responses classified into one of those 4 groups.

Proportion of working years at this school (PROPWORK_Q)

In TC007 teachers were asked to how many years of teaching experience they had at the school having them take the questionnaire and how many years of teaching experience they had in total. These responses were used to compute the proportion of years the teacher has been working in the school, by dividing the number of years at their current school by total number of years teaching. The international distribution of these proportions was divided into four equal groups (quartiles). The PROPWORK_Q index contains principal’s responses classified into one of those 4 groups.

Derived variables based on IRT Scaling

The Teacher Questionnaire provided data for 23 DVs based on IRT scaling. The Cronbach’s alpha for each scale and group are presented in Annex Table 22.A.102 the number of items with international parameters for each scale and group are presented in Table 22.A.103, the number of trend items with international parameters for each trend scale and group are presented in Annex Table 22.A.104, and the countries/economies for which the scale scores were suppressed for each scale are presented in Annex Table 22.A.105.

Exchange and co-ordination for teaching (EXCHT)

Teachers’ frequency ratings of how often they participate in teaching-related co-operation (e.g., Exchange teaching materials with colleagues”, “Attend team conferences”) in question TC046 were scaled into the index of “exchange and co-ordination for teaching”, which is linked to the PISA 2018 scale EXCHT through anchoring the parameters for four link items. Each of the five items included in this scale had six response options (“Never”, “Once a year or less”, “2-4 times a year”, “5-10 times a year”, “1-3 times a month”, “Once a week or more”). Annex Table 22.A.106 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Need for professional development (DEVNEED)

Teachers’ ratings of their need for professional development in various areas (e.g., “knowledge of the curriculum”, “student behaviour and classroom management”) in question TC185 were scaled into the index of “need for professional development”, which is linked to the PISA 2022 scale DEVNEED through anchoring the parameters for 11 link items included in this scale. Each of the 15 items included in this scale had four response options (“no need at present”, “low level of need”, “moderate level of need”, “high level of need”). Annex Table 22.A.107 shows the item wording and item parameters for the items in this scale,

and it indicates which items were link items. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teacher's self-efficacy in classroom management (SEFFCM)

Teachers' ratings of their classroom management skills (e.g., "Control disruptive behaviour in the classroom", "Get students to follow classroom rules") in question TC199 were scaled into the index of "teacher's self-efficacy in classroom management", which is linked to the SEFFCM scale in PISA 2018 through anchoring parameters for the four link items. Each of the four items included in this scale had four response options ("not at all", "to some extent", "quite a bit", "a lot"). Annex Table 22.A.108 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teacher's self-efficacy in instructional settings (SEFFINS)

Teachers' ratings of their self-efficacy in instructional settings (e.g., "craft good questions for my students", "provide an alternative explanation for example when students are confused") in question TC199 were scaled into the index of "teacher's self-efficacy in instructional settings", which is linked to the SEFFINS scale in PISA 2018 through anchoring parameters for the four included link items. Each of the six items included in this scale had four response options ("not at all", "to some extent", "quite a bit", "a lot"). Annex Table 22.A.109 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teacher's self-efficacy in maintaining positive relations with students (SEFFREL)

Teachers' ratings of their ability to maintain positive relations with students (e.g., "help my students value learning", "motivate students who show low interest in school work") in question TC199 were scaled into the index of "teacher's self-efficacy in maintaining positive relations with students, which is linked to the SEFFREL scale in PISA 2018. Each of the four items included in this scale had four response options ("not at all", "to some extent", "quite a bit", "a lot"). Annex Table 22.A.110 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teacher use of ICT (TCDIGRES)

Teachers' frequency ratings of how often they use ICT for various teaching tasks (e.g., "use <digital resources> to design tasks", "use <digital resources> to provide feedback to students") in question TC220 were scaled into the index of "teacher use of ICT", which is linked to the PISA 2022 scale TCDIGRES through anchoring parameters for four link items. Each of the six items included in this scale had five response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day"). Annex Table 22.A.111 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teachers' self-efficacy related to teaching science content (SETEACH)

Teachers were asked to which extent they could do different activities related to the teaching of science (e.g., “explain a complex scientific concept to a fellow teacher”, “explain the links between biology, physics and chemistry”) in question TC033 were scaled into the index of “teachers' self-efficacy related to teaching science content”, which is linked to PISA 2015 scale SETEACH through anchoring of the four included link items. Each of the four items included in this scale had four response options (“not at all”, “very little”, “to some extent”, “to a large extent”). Annex Table 22.A.112 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Learning about cooperation with parents/guardians (PARCOOPLRN)

Teachers were asked how much they had learned about co-operation between parents or guardian and teachers in different contexts (e.g., “during teacher training as a specific topic”, “during professional development workshops”). Five items included in in question TC409 with four response options (“not at all”, “very little”, “to some extent”, “to a large extent”) were used to derive the new PISA 2025 index “learning about cooperation with parents/guardians” (PARCOOPLRN). Annex Table 22.A.113 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Positive leadership perceptions (POSLEAD)

Teachers' ratings of their agreement with statements about leadership at their school (e.g., “I enjoy projects that require creative solutions.”, “I express myself through art.”) in question TC407 were scaled into the new PISA 2025 index “positive leadership perceptions” (POSLEAD). Each of the seven items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.114 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teachers' enjoyment of broad science (ENJBSCIE)

Teachers' ratings of their agreement with statements about their enjoyment of broad science (e.g., “I enjoy learning <broad science> topics”, “I enjoy acquiring new knowledge in <broad science>”) in question TC419 were scaled into the new PISA 2025 index of “teachers' enjoyment of broad science” (ENJBSCIE). Each of the four items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.115 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teachers' positive perceptions of digital resources (PERSDIGRES)

Teachers' ratings of their agreement with statements about advantages of using digital resources in the teaching of science (e.g., “<digital resources> increase student engagement”, “<Digital resources> promote collaboration among students”) in question TC426 were scaled into the new PISA 2025 index of “teachers' positive perceptions of digital resources” (PERSDIGRES). Each of the seven items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.116 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Science activities with students (STSCIACT)

Teachers were asked to indicate the frequency of undertaking different activities to improve students' understanding of science (e.g., "organise visits with scientists to work with students", "run a science club outside of set class times") in question TC408 were scaled into the new PISA 2025 index of "science activities with students" (STSCIACT). Each of the five items included in this scale had four response options ("never", "rarely", "sometimes", "frequently"). Annex Table 22.A.117 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Use of formative assessments (USEFORAS)

Teachers were asked to indicate the frequency of using formative assessment to evaluate students' learning in school science (e.g., "I have individual students answer questions in front of the class", "I collect data from classroom assignments or homework") in question TC415 were scaled into the new PISA 2025 index of "use of formative assessments" (USEFORAS). Each of the nine items included in this scale had four response options ("never or almost never", "some lessons", "many lessons", "every or almost every lesson"). Annex Table 22.A.118 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teachers' encouragement of students (ENCOURSTUD)

Teachers were asked to indicate the frequency of using different ways of encouraging student learning in science classes (e.g., "tell students that making mistakes is part of the learning process", "ask students to develop learning goals for science") in question TC417 were scaled into the new PISA 2025 index of "teachers' encouragement of students" (ENCOURSTUD). Each of eight items included in this scale had four response options ("never", "rarely", "sometimes", "frequently"). Annex Table 22.A.119 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Teachers' confidence in explaining science content (CONEXSCI)

Teachers were asked about the extent to which they could do different things when explaining science content (e.g., "explain a complex scientific concept to a fellow teacher", "explain the links between biology, physics and chemistry"). Four items included in in question TC034 with four response options ("not at all", "very little", "to some extent", "to a large extent") were used to derive the new PISA 2025 index "teachers' confidence in explaining science content" (CONEXSCI). Annex Table 22.A.120 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Digital resource use in science teaching (DRUSESCT)

Teachers' frequency ratings of how often they instructed students to do tasks using(e.g., "use <digital resources> to make students observe a natural system or phenomenon", "use <digital resources> to collect, record, and analyse data") in question TC428 were scaled into the index of "Negative physical symptoms". Each of the 10 items included in this scale had five response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day"). Annex Table 22.A.121 shows the item wording and item parameters for the items in this scale. This scale was also derived from the FLA teacher questionnaire using anchored parameters and set to the same metric.

Disciplinary climate in science (TCDISCLISC)

Teachers were asked to indicate the frequency with which certain things happened in class that reflect the disciplinary climate in science lessons (e.g., “there is noise and disorder”, “students don’t start working for a long time after the lesson begins”) in question TC414 were used to derive the new PISA 2025 index “disciplinary climate in science” (TCDISCLISC). Each of the eight items included in this scale had four response options (“every lesson”, “most lessons”, “some lessons”, “never or almost never”). The items were not reverse-coded so that higher scale scores reflect a more positive disciplinary climate in science lessons. Annex Table 22.A.122 shows the item wording and item parameters for the items in this scale.

Teachers’ beliefs concerning learning growth (BELLRNG)

Teachers’ ratings of their agreement with statements about beliefs concerning their students’ learning growth (e.g., “my students can all reach their goals with the right amount of support”, “my students feel comfortable taking risks”) in question TC418 were used to derive the new PISA 2025 index of “teachers’ beliefs concerning learning growth” (BELLRNG). Each of the four items included in this scale had four response options (“strongly agree”, “agree”, “disagree”, “strongly disagree”) and all items were reverse-coded so that higher scale scores indicate higher levels of beliefs in learning growth. Annex Table 22.A.123 shows the item wording and item parameters for the items in this scale.

Science teacher collaboration (SCIETCOLL)

Teachers’ frequency ratings of how often they collaborate with fellow school science teachers (e.g., “we exchange tasks for lessons and homework that cover a range of different levels of difficulty”, “we discuss ways to teach learning strategies and techniques to our students”) in question TC411 were used to derive the new PISA 2025 index “science teacher collaboration” (SCIETCOLL). Each of the nine items included in this scale had four response options (“never”, “rarely”, “sometimes”, “frequently”). Annex Table 22.A.124 shows the item wording and item parameters for the items in this scale.

Parents/guardians’ involvement in science learning (PARSCLRN)

Teachers’ frequency ratings of how often parents or guardians are involved in situations concerning students’ science learning (e.g., “discussing their child’s science academic performance with you”, “discussing concerns with science curriculum content”) in question TC410 were used to derive the new PISA 2025 index “parents/guardians’ involvement in science learning” (PARSCLRN). Each of the six items included in this scale had four response options (“never”, “rarely”, “sometimes”, “frequently”). Annex Table 22.A.125 shows the item wording and item parameters for the items in this scale.

Teachers’ beliefs about the nature of science (TCBELSCIE)

Teachers’ ratings of their agreement with statements about beliefs concerning their beliefs about the nature of science (e.g., “good answers are based on evidence from many different experiments”, “information from one source should be checked from a range of other reliable sources before it is accepted”) in question TC420 were used to derive the new PISA 2025 index of “teachers’ beliefs about the nature of science” (TCBELSCIE). Each of the eight items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.126 shows the item wording and item parameters for the items in this scale.

Teachers’ participation in environment-related activities (TCPENVACT)

Teachers were asked to indicate the frequency of participating in different environmental actions (e.g., “joined a peaceful march or rally related to an environmental issue”, “posted concerns about an

environmental issue on social media”) in question TC424 were used to derive the new PISA 2025 index “teachers’ participation in environment-related activities” (TCPENVACT). Each of the five items included in this scale had four response options (“not at all”, “sometimes”, “frequently”, “very frequently”) and higher scale scores indicate higher levels of teacher participation. Annex Table 22.A.127 shows the item wording and item parameters for the items in this scale.

Teachers’ environmental awareness (TCENWARE)

Teachers were asked to report how informed they felt about different environmental issues (e.g., “the increase of greenhouse gases in the atmosphere”, “the consequences of clearing forests for other land use”). The 14 items included in question TC421 had four response options (“I have never heard of this”, “I have heard about this but I would not be able to explain what it is really about”, “I know something about this and could explain the general issue”, “I am familiar with this and I would be able to explain this well”) and were used to derive the index “teachers’ environmental awareness” (TCENWARE), for which higher scale scores indicate higher levels of environmental awareness. Annex Table 22.A.128 shows the item wording and item parameters for the items in this scale.

FLA Student Questionnaire indices

There were 34 indices that were derived from the FLA student questionnaire, including 28 simple DVs, six IRT-scaled DVs, and one complex composite index (ESCS). The DVs are shown in Annex Table 22.A.129 and will be described in the following section. This section only describes those indices that were specific to the FLA Student questionnaire.

Simple questionnaire indices

Student background, environment and family support

English language proficiency parent/guardian 1 (PT1ENG)

Students responded to a question about whether their parent/guardian 1 could read or write in English (ST406 or ST411, depending on the branch students followed based on responses to 403 – number of parents/guardians). PT1ENG takes a value of “0” if parent/guardian 1 cannot read nor write in English, or if student doesn’t know, a value of “1” if parent/guardian 1 can only read in English and “2” if parent/guardian 1 can write in English. The index takes a value of 3 if parent/guardian 1 can read and write in English.

English language proficiency parent/guardian 2 (PT2ENG)

Students responded to a question about whether their parent/guardian 2 could read or write in English (ST412, only for students who followed the two parents/guardians branch based on responses to 403 – number of parents/guardians). PT2ENG takes a value of “0” if parent/guardian 2 cannot read nor write in English, or if student doesn’t know, a value of “1” if parent/guardian 2 can only read in English and “2” if parent/guardian 1 can write in English. The index takes a value of 3 if parent/guardian 2 can read and write in English.

Language spoken at home (LANGNFL)

Students indicated what language they usually spoke at home (ST022FLQ01DA), and the database includes a variable (LANGNFL) containing country/economy-specific code for each language.

The setting for language setting at school

Starting age for English studies (ENLRNAGE)

Students' answers on question ST202FL about how old they were when they had their first English lesson at school were recoded into the ENLRNAGE index to reflect numeric values. ENLRNAGE ranges from 5 to 15.

Number of lessons in foreign languages (LANGLESS)

Students responded to a question about the languages in which they took lessons at school (ST157FL). LANGLESS counts the number of languages students selected. The index takes the value of 0 if the student did not select any language in ST157FL and a value of 10 if the student selected all nine languages presented in the question and also "Other language".

Number of subjects taught in English (SUBJSENG)

Students responded to a question about whether any of some specific lessons they have at school were taught in English (ST151FL). SUBJSENG counts the number of lessons students selected. The index takes the value of 0 if the student did not select any lesson in ST151FL and a value of 8 if the student selected all seven lessons presented in the question and also "My lessons in other subjects are taught in <English>".

Derived variables based on IRT Scaling

The FLA Student Questionnaire provided data for three FLA-specific DVs based on IRT scaling in addition to five IRT-scaled DVs that were also included in the core Student Questionnaire. The Cronbach's alpha for each scale and group are presented in Annex Table 22.A.130, the number of items with international parameters for each scale and group are presented in Annex Table 22. A.131, the number of trend items with international parameters for each trend scale and group are presented in Annex Table 22. A.132, the countries/economies for which the scale scores were suppressed for each scale are presented in Annex Table 22. A.133, and the groups that did not administer each scale are presented in Annex Table 22. A.134.

Student background, environment and family support

Heritage language proficiency (HLANGPROF)

Students were asked to rate how they felt about their ability to do different things in another language spoken at home apart from their language used in the PISA assessment (e.g., "speak with friends or family", "read about an event (e.g. a festival or a sport match) from a website or newspaper"). The five items included in question ST480 with four response options ("not at all", "not very confident", "confident", "very confident") were used to derive the new PISA 2025 index "heritage language proficiency" (HLANGPROF), for which higher scale score indicate higher levels of confidence. Annex Table 22. A.135 shows the item wording and item parameters for the items in this scale.

Learners' attitudes, motivations and behaviours

Anxiety towards English language learning (ANXENGL)

Students were asked to rate their agreement with statements about their feelings towards English language learning (e.g., "I am afraid to make grammar mistakes when speaking English", "I get nervous when I am

listening to someone speaking English”). Seven out of 10 items included in question ST247 with four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”) were used to derive the new PISA 2025 index “anxiety towards English language learning” (ANXENGL), for which higher scale score indicate higher levels of anxiety towards English language learning. Annex Table 22. A.136 shows the item wording and item parameters for the items in this scale.

Expectations to use English in the future (EXPENGUSE)

Students were asked to which extent they expected to use English for different purposes in the future (e.g., “interact with locals in English while travelling”, “move abroad to work or study in English”). The six items included in question ST135 with four response options (“not at all”, “very little”, “to some extent”, “definitely”) were used to derive the new PISA 2025 index “expectations to use English in the future (EXPENGUSE), for which higher scale score indicate higher levels of expectation to use English. Annex Table 22. A.137 shows the item wording and item parameters for the items in this scale.

FLA School Questionnaire indices

The FLA school questionnaire included one simple FLA-specific DV but no FLA-specific scaled DVs. The DV is shown in Annex Table 22. A.138 and will be described in the following section.

Simple questionnaire indices

The setting for language setting at school

English class size at school (ENCLSIZE)

Principals were asked in SC208FL about the average size of the test language classes in their school. The nine response categories were “15 students or fewer”, “16-20 students”, “21-25 students”, “26-30 students”, “31-35 students”, “36-40 students”, “41-45 students”, “46-50 students”, and “More than 50 students”. The average class size (ENCLSIZE) was derived from the midpoint of each response category, resulting in a value of 13 for the lowest category, and a value of 53 for the highest.

FLA Parent Questionnaire indices

The FLA Parent Questionnaire is an international option that countries/economies participating the Foreign Language Assessment (FLA) could choose to implement. There were 11 indices that were derived from the FLA parent questionnaire, including two simple DVs and nine IRT-scaled DVs. The DVs are shown in Annex Table 22.A.139 and will be described in the following section.

Simple questionnaire indices

Student background, environment and family support

Number of home languages (NHOMLANG)

Parents/guardians were asked to indicate the languages in which the people they live with speak at home (ST219FL). The index NHOMLANG takes the value of “1” when only one language was selected from the list of those presented in the question, a value of “2” when two languages were selected, and a value of “3” when 3 or more languages were selected.

Derived variables based on IRT Scaling

The FLA Parent Questionnaire provided data for five FLA-specific DVs based on IRT scaling in addition to three IRT-scaled DVs that were also included in the core Parent Questionnaire. The Cronbach's alpha for each scale and group are presented in Annex Table 22.A.140, the number of items with international parameters for each scale and group are presented in Annex Table 22.A.141, and the number of trend items with international parameters for each trend scale and group are presented in Annex Table 22.A.142.

Student background, environment and family support

Family support for English learning (FENGSUP)

Parents/guardians were asked to indicate how often they or someone else living in the home undertook activities to support the child's English learning (e.g., "help my child with English homework", "help my child prepare for English school tests"). The five items included in question PA230 with five response options ("never or almost never", "about once or twice a year", "about once or twice a month", "about once or twice a week", "every day or almost every day") were used to derive the new PISA 2025 index "family support for English learning" (FENGSUP), for which higher scale score indicate higher levels of support. Annex Table 22.A.143 shows the item wording and item parameters for the items in this scale.

Parental satisfaction with English learning at school (PARSAT)

Parents/guardians were asked to rate their agreement with statements about their satisfaction with English learning at their child's school (e.g., "I am satisfied with how <English'> is taught at my child's school", "I am satisfied with the quality of <English'> lessons my child receives at school"). The three items in question PA236 with four response options ("strongly disagree", "disagree", "agree", "strongly agree") were used to derive the new PISA 2025 index "Parental satisfaction with English learning at school" (PARSAT), for which higher scale score indicate higher levels of satisfaction. Annex Table 22.A.144 shows the item wording and item parameters for the items in this scale.

Parental perception of the importance of learning English (PARIMPO)

Parents/guardians were asked about their opinions about the importance of learning English for different purposes (e.g., "for getting a good job", "for using computers and other technical devices"). The five items included in question PA295 had four response options ("not important at all", "not very important", "important", "very important") and were used to derive the new PISA 2025 index "parental perception of the importance of learning English" (PARIMPO), for which higher scale score indicate higher levels of perceived importance. Annex Table 22.A.145 shows the item wording and item parameters for the items in this scale.

Parental perceptions of own reading proficiency in English (PARENGL)

Parents/guardians were asked about extent to which they could do different things when speaking or reading English (e.g., "give a short description of very familiar topics (e.g. yourself, your family)", "understand words commonly used in everyday life (e.g. common street signs, posters, menus)."). The five items included in question PA225 and PA228 had four response options ("not at all", "very little", "to some extent", "to a large extent") and were used to derive the new PISA 2025 index "parental perceptions of own reading proficiency in English" (PARENGL), for which higher scale score indicate higher levels of perceived proficiency. Annex Table 22.A.146 shows the item wording and item parameters for the items in this scale.

Parental use of English at work (PARUSE)

Parents/guardians were asked to indicate how often they used English at work during the last twelve months (e.g., “speak English (e.g. giving presentations, participating in conversations)”, “write in English (e.g. writing emails, drafting reports)”). The four items included in question PA264 had five response options (“never or almost never”, “about once or twice a year”, “about once or twice a month”, “about once or twice a week”, “every day or almost every day”) and were used to derive the new PISA 2025 index “parental use of English at work” (PARUSE), for which higher scale score indicate higher levels of English use at work. Annex Table 22.A.147 shows the item wording and item parameters for the items in this scale.

FLA Teacher Questionnaire indices

The FLA Teacher Questionnaire is an international option that countries/economies participating the Foreign Language Assessment (FLA) could choose to implement. There were 45 variables derived from this questionnaire, including 32 simple DVs and 13 IRT scaled DVs. An overview of all DVs in this questionnaire is shown in Annex Table 22.A.148 and each are described in the following sections. This section only describes those indices that were specific to FLA while common indices are described in the section of the core Science teacher Questionnaire.

The FLA Teacher Questionnaire only included simple DVs that were also included in core Teacher Questionnaire and are described in the corresponding section.

Derived variables based on IRT Scaling

The FLA Teacher Questionnaire provided data for a total of 13 DVs based on IRT scaling, five of which are specific to FLA and described in this section. The Cronbach’s alpha for each scale and group are presented in Annex Table 22.A.149, the number of items with international parameters for each scale and group are presented in Annex Table 22.A.150, the number of trend items with international parameters for each trend scale and group are presented in Annex Table 22.A.151, and the countries/economies for which the scale scores were suppressed for each scale are presented in Annex Table 22.A.152.

The Teacher

Job satisfaction among English teachers (JOBSATENG)

Teachers’ ratings of their agreement with different statements reflecting satisfaction with their job as English teachers (e.g., “I enjoy teaching English”, “if I could choose again, I would still choose to become an English teacher”) in question TC114 were used to derive the new PISA 2025 index of “job satisfaction among English teachers” (JOBSATENG). Each of six four items included in this scale had four response options (“strongly disagree”, “disagree”, “agree”, “strongly agree”). Annex Table 22.A.153 shows the item wording and item parameters for the items in this scale.

Teacher self-efficacy for improving students’ attitudes towards English (TCSELFATT)

Teachers’ ratings of their ability to improve students’ attitudes towards English (e.g., “help students communicate in English”, “help students overcome their anxiety towards communicating in English”) in question TC115 were used to derive the new PISA 2025 index of “teacher self-efficacy for improving students’ attitudes towards English (TCSELFATT). Each of the five items included in this scale had four response options (“Not at all”, “very little”, “to some extent”, “a lot”). Annex Table 22.A.154 shows the item wording and item parameters for the items in this scale.

Teacher self-efficacy for using English in the classroom (TCSELFUSE)

Teachers' ratings of their ability to use English in the classroom for different tasks (e.g., "explain tasks in English", "give students oral feedback in English") in question TC190 were used to derive the new PISA 2025 index of "teacher self-efficacy for using English and motivating students in the classroom" (TCSELFUSE). Each of the six items included in this scale had four response options ("Not at all", "very little", "to some extent", "a lot"). Annex Table 22.A.155 shows the item wording and item parameters for the items in this scale.

Teacher use of English outside the classroom (TCUSEENG)

Teachers were asked to indicate the frequency of using English to communicate in different situations outside the classroom during the last 12 months (e.g., "face-to-face communication with my friends in my country", "remote spoken interaction (e.g. by phone or conference call)") in question TC111 were used to derive the new PISA 2025 index of "teacher use of English outside the classroom" (TCUSEENG). Each of the six items included in this scale had five response options ("never", "once or twice", "about once or twice a month", "about once or twice a week", "every or almost every day"). Annex Table 22.A.156 shows the item wording and item parameters for the items in this scale.

Assessment practices

Use of formative assessments in English lessons (TCENGFORM)

Teachers were asked to indicate the frequency of using formative assessment practices in English lessons (e.g., "I give students feedback on their strengths", "I tell students which areas they can improve on") in question TC302 were used to derive the new PISA 2025 index of "use of formative assessments in English lessons" (TCENGFORM). Each of the six items included in this scale had five response options ("never or almost never", "less than half of the lessons", "about half of the lessons", "more than half of the lessons", "every or almost every lesson"). Annex Table 22.A.157 shows the item wording and item parameters for the items in this scale.

References

- Adams, R. J. (2020). *ACER ConQuest: Generalised Item Response Modelling Software (Version 5) [Computer software]*. Educational Testing Service.
- Adams, R. J., Wu, M. L., Cloney, D., Berezner, A., & Wilson, M. (2020). *ACER ConQuest: Generalised Item Response Modelling Software*. Camberwell, Victoria: Australian Council for Educational Research.
- Avvisati, F. (2020). The measure of socio-economic status in PISA: A review and some suggested improvements. *Large-Scale Assessments in Education*, 8(1), 1-37. doi:10.1186/s40536-020-00086-x
- Bertling, J. P., & Weeks, J. P. (2018). *Plans for Within-construct Questionnaire Matrix Sampling in PISA 2021*. Paper presented to PISA Technical Advisory Group, August 2018, Princeton, NJ.
- Bertling, J. P., & Weeks, J. P. (2020). *Getting More Bang for Your Buck: Within-construct Questionnaire Matrix Sampling*. Paper presented to PISA Technical Advisory Group, September 2020, Princeton, NJ.
- Bertling, J. P., Weeks, J. P., Kandathil, M., Kapur, L., Zhang, S., & Zhao, W. (2020). Comparison of within-construct matrix sampling with scale shortening: Impact on IRT scaling and population modelling. In *Memo prepared for PISA Technical Advisory Group, TAG (2020)2*.
- Birnbaum, A. (1968). Some latent trait models and their use in inferring an examinee's ability. In F. M. Lord, & M. R. Novick (Eds.), *Statistical Theories of Mental Test Scores* (pp. 397-479). Addison-Wesley.
- Cowan, C. D., Hauser, R. M., Kominski, R. A., Levin, H. M., Lucas, S. R., Morgan, S. L., . . . Chapman, C. (2012). *Improving the measurement of socioeconomic status for the National Assessment of Educational Progress: A theoretical foundation*. National Center for Education Statistics. Retrieved from https://nces.ed.gov/nationsreportcard/pdf/researchcenter/socioeconomic_factors.pdf
- Ganzeboom, H. B., & Treiman, D. J. (2003). Three internationally standardised measures for comparative research on occupational status. In J. H. Hoffmeyer-Zlotnik, & C. Wolf (Eds.), *Advances in Cross-National Comparison: A European Working Book for Demographic and Socio-Economic Variables* (pp. 159-193). Kluwer Academic Press.
- Masters, G. N. (1982). A Rasch model for partial credit scoring. *Psychometrika*, 47(2), 149-174. doi:10.1007/BF02296272
- Muraki, E. (1992). A generalized partial credit model: Application of an EM algorithm. *Applied Psychological Measurement*, 16(2), 159-177. doi:10.1002/j.2333-8504.1992.tb01436.x
- OECD. (2020). *PISA 2018 Technical Report*. PISA, OECD Publishing, Paris. Retrieved from <https://www.oecd.org/en/about/programmes/pisa/pisa-data.html>
- OECD. (2024). *PISA 2022 Technical Report*. Paris: OECD Publishing. doi:<https://doi.org/10.1787/01820d6d-en>
- OECD. (2026). *PISA 2025 Assessment and Analytical Framework*. Paris: OECD Publishing. doi:<https://doi.org/10.1787/86c36975-en>
- OECD/Eurostat/UNESCO Institute for Statistics. (2015). *ISCED 2011 Operational Manual: Guidelines for Classifying National Education Programmes and Related Qualifications*. OECD Publishing, Paris. doi:10.1787/9789264228368-en
- Rasch, G. (1960). *Probabilistic Models for Some Intelligence and Attainment Tests*. Nielsen and Lydiche.
- von Davier, M. (2015). Multidimensional discrete latent trait models (mdltn) (Version 1.965) [Computer software].
- Warm, T. A. (1989). Weighted likelihood estimation of ability in item response theory. *Psychometrika*,

Notes

1. For the seven trend scales linked to PISA 2012, the Rasch model (Rasch, 1960) was used to scale the dichotomous items, while the partial credit model (PCM) was used to scale the polytomous items, in line with the models used in PISA 2012.
2. The International Standard Classification of Education (ISCED) is used in international educational statistics to classify levels in education systems worldwide. A link to the 2011 framework, ISCED 2011, used in PISA 2025 can be found at <https://uis.unesco.org/sites/default/files/documents/international-standard-classification-of-education-isced-2011-en.pdf>
3. Separate simple DVs (ST251D08JA, ST251D09JA) combine all the customizations across countries/economies to ST251Q08JA, and ST251Q09JA, respectively. See Annex D.
4. The random value was drawn from a normal distribution with a mean of 0 and a standard deviation equal to the standard deviation of the residuals of the regression model for the country/economy.
5. Due to missing data within the OECD countries, the senate weights of the individuals that were ultimately included in the calculation of the mean and standard deviation of each component did not sum to 5,000 for each OECD country. Therefore, the OECD countries were only approximately equally weighted in the calculation of the mean and standard deviation of each component.
6. Again, due to missing data within the OECD countries, the senate weights of the individuals that were ultimately included in the calculation of the mean and standard deviation of the preliminary ESCS scores did not sum to 5,000 for each OECD country. Therefore, the OECD countries were only approximately equally weighted in the calculation of the mean and standard deviation of the preliminary ESCS scores.

Annex 22.A. Comprehensive Scales and Variables Analysis from PISA 2025 Questionnaires

List of Tables in Annex 22.A

All tables are available online in the Excel file *22-AnnexTables.xlsx*.

Table 22.A.1	List of all trend scales
Table 22.A.2	OECD mean and standard deviation of the original WLEs for the new scales
Table 22.A.3	OECD mean and standard deviation of the original WLEs in PISA 2022 for trend scales linked to PISA 2025
Table 22.A.4	OECD mean and standard deviation of the original WLEs in PISA 2018 for trend scales linked to PISA 2025
Table 22.A.5	OECD mean and standard deviation of the original WLEs in PISA 2012 for trend scales linked to PISA 2025
Table 22.A.6	Scaling constants used as linear transformations for original WLEs in PISA 2025 to link trend scales to PISA 2015 (and PISA 2006)
Table 22.A.7	Variables derived from the Student Questionnaire
Table 22.A.8	Mapping of ISCED levels to years of education
Table 22.A.9	ISCEDP values and labels
Table 22.A.10	Creation of EXPECEDU index
Table 22.A.11	Cronbach's alpha for the IRT scales in the Student Questionnaire
Table 22.A.12	Number of items with international parameters for the IRT scales in the Student Questionnaire
Table 22.A.13	Number of trend items with international parameters for the trend IRT scales in the Student Questionnaire
Table 22.A.14	Countries/economies for which the scale scores were suppressed for the IRT scales in the Student Questionnaire
Table 22.A.15	Groups that did not administer the IRT scales in the Student Questionnaire
Table 22.A.16	Items in the HOMEPOS scale
Table 22.A.17	Recoding of items in the HOMEPOS scale
Table 22.A.18	Items in the ICTRES scale
Table 22.A.19	Items in the BULLIED scale
Table 22.A.20	Items in the FEELSAFE scale
Table 22.A.21	Items in the BELONG scale
Table 22.A.22	Items in the COLLENEFF scale
Table 22.A.23	Items in the JOYSCIE scale
Table 22.A.24	Items in the OPENVLRN scale
Table 22.A.25	Items in the EFFSCIE scale
Table 22.A.26	Items in the STBELSCIE scale
Table 22.A.27	Items in the ENVCAPCH scale
Table 22.A.28	Items in the ENVAWARE scale
Table 22.A.29	Items in the ENVAPART scale
Table 22.A.30	Items in the CURIO scale
Table 22.A.31	Items in the PERSEV scale
Table 22.A.32	Items in the ENGSCIPR scale
Table 22.A.33	Items in the COGACSC scale
Table 22.A.34	Items in the SCIECOLLR scale

Table 22.A.35	Items in the DISCLISCI scale
Table 22.A.36	Items in the TEACHSUP scale
Table 22.A.37	Items in the FAMSUP scale
Table 22.A.38	Items in the AIUESCH scale
Table 22.A.39	Items in the COGABIL scale
Table 22.A.40	Items in the ENPROBS scale
Table 22.A.41	Items in the GOALSET scale
Table 22.A.42	Items in the SELFREG scale
Table 22.A.43	OECD mean and standard deviation of each component of ESCS and the preliminary ESCS scores for PISA 2025
Table 22.A.44	Variables derived from the ICT Familiarity Questionnaire
Table 22.A.45	Cronbach's alpha for the IRT scales in the ICT Familiarity Questionnaire
Table 22.A.46	Number of items with international parameters for the IRT scales in the ICT Familiarity Questionnaire
Table 22.A.47	Items in the ICTSCH scale
Table 22.A.48	Items in the ICTHOME scale
Table 22.A.49	Items in the ICTQUAL scale
Table 22.A.50	Items in the ICTFEED scale
Table 22.A.51	Items in the ICTOUT scale
Table 22.A.52	Items in the ICTWKDY scale
Table 22.A.53	Items in the ICTWKEND scale
Table 22.A.54	Items in the ICTREG scale
Table 22.A.55	Items in the ICTINFO scale
Table 22.A.56	Items in the DIGINTLRN scale
Table 22.A.57	Items in the INTICT scale
Table 22.A.58	Items in the SOIACT scale
Table 22.A.59	Variables derived from the Parent Questionnaire
Table 22.A.60	Creation of the PAREXPT index
Table 22.A.61	Cronbach's alpha for the IRT scales in the Parent Questionnaire
Table 22.A.62	Number of items with international parameters for the IRT scales in the Parent Questionnaire
Table 22.A.63	Number of trend items with international parameters for the trend IRT scales in the Parent Questionnaire
Table 22.A.64	Items in the PRESUPP scale
Table 22.A.65	Items in the FGENSUP scale
Table 22.A.66	Items in the FSCISUP scale
Table 22.A.67	Items in the PARINVOL scale
Table 22.A.68	Items in the PVDRSCUSE scale
Table 22.A.69	Items in the PQENPERC scale
Table 22.A.70	Items in the PQSELFREG scale
Table 22.A.71	Items in the PQGENSCI scale
Table 22.A.72	Items in the PQENVOPT scale
Table 22.A.73	Items in the PQDIGEFF scale
Table 22.A.74	Items in the PQFEED scale
Table 22.A.75	Items in the PQSCHOOL scale
Table 22.A.76	Items in the PQSCAR scale
Table 22.A.77	Variables derived from the School Questionnaire
Table 22.A.78	Cronbach's alpha for the IRT scales in the School Questionnaire
Table 22.A.79	Number of items with international parameters for the IRT scales in the School Questionnaire
Table 22.A.80	Number of trend items with international parameters for the trend IRT scales in the School Questionnaire
Table 22.A.81	Countries/economies for which the scale scores were suppressed for the IRT scales in the School Questionnaire
Table 22.A.82	Groups that did not administer the IRT scales in the School Questionnaire
Table 22.A.83	Items in the PISASUPP scale
Table 22.A.84	Items in the NEGSCIM scale

Table 22.A.85	Items in the EDUSHORT scale
Table 22.A.86	Items in the STAFFSHORT scale
Table 22.A.87	Items in the OPPACTENV scale
Table 22.A.88	Items in the ALLACTIV scale
Table 22.A.89	Items in the AVLRSOFT scale
Table 22.A.90	Items in the DIGPREP scale
Table 22.A.91	Items in the EDULEAD scale
Table 22.A.92	Items in the DELGLOB scale
Table 22.A.93	Items in the PRIORLRN scale
Table 22.A.94	Items in the FREESCIE scale
Table 22.A.95	Items in the SCTRAIN scale
Table 22.A.96	Items in the SESASSESS scale
Table 22.A.97	Items in the STDTEST scale
Table 22.A.98	Items in the TDTEST scale
Table 22.A.99	Items in the ENCOURPG scale
Table 22.A.100	Variables derived from the Teacher Questionnaire
Table 22.A.101	Creation of the TCISCED index
Table 22.A.102	Cronbach's alpha for the IRT scales in the Teacher Questionnaire
Table 22.A.103	Number of items with international parameters for the IRT scales in the Teacher Questionnaire
Table 22.A.104	Number of trend items with international parameters for the trend IRT scales in the Teacher Questionnaire
Table 22.A.105	Countries/economies for which the scale scores were suppressed for the IRT scales in the Teacher Questionnaire
Table 22.A.106	Items in the EXCHT scale
Table 22.A.107	Items in the DEVNEED scale
Table 22.A.108	Items in the SEFFCM scale
Table 22.A.109	Items in the SEFFINS scale
Table 22.A.110	Items in the SEFFREL scale
Table 22.A.111	Items in the TCDIGRES scale
Table 22.A.112	Items in the SETEACH scale
Table 22.A.113	Items in the PARCOOPLRN scale
Table 22.A.114	Items in the POSLEAD scale
Table 22.A.115	Items in the ENJBSCIE scale
Table 22.A.116	Items in the PERSDIGRES scale
Table 22.A.117	Items in the STSCIACT scale
Table 22.A.118	Items in the USEFORAS scale
Table 22.A.119	Items in the ENCOURSTUD scale
Table 22.A.120	Items in the CONEXSCI scale
Table 22.A.121	Items in the DRUSESCT scale
Table 22.A.122	Items in the TCDISCLISC scale
Table 22.A.123	Items in the BELLRNG scale
Table 22.A.124	Items in the SCIETCOLL scale
Table 22.A.125	Items in the PARSCLRN scale
Table 22.A.126	Items in the TCBELSCIE scale
Table 22.A.127	Items in the TCPENVACT scale
Table 22.A.128	Items in the TCENWARE scale
Table 22.A.129	Variables derived from the FLA Student Questionnaire (FLA-specific indices only)
Table 22.A.130	Cronbach's alpha for the IRT scales in the FLA Student Questionnaire
Table 22.A.131	Number of items with international parameters for the IRT scales in the FLA Student Questionnaire
Table 22.A.132	Number of trend items with international parameters for the trend IRT scales in the FLA Student Questionnaire
Table 22.A.133	Countries/economies for which the scale scores were suppressed for the IRT scales in the FLA Student Questionnaire

Table 22.A.134	Groups that did not administer the IRT scales in the FLA Student Questionnaire
Table 22.A.135	Items in the HLANGPROF scale
Table 22.A.136	Items in the ANXENGL scale
Table 22.A.137	Items in the EXPENGUSE scale
Table 22.A.138	Variables derived from the FLA School Questionnaire (FLA-specific indices only)
Table 22.A.139	Variables derived from the FLA Parent Questionnaire (FLA-specific indices only)
Table 22.A.140	Cronbach's alpha for the IRT scales in the FLA Parent Questionnaire
Table 22.A.141	Number of items with international parameters for the IRT scales in the FLA Parent Questionnaire
Table 22.A.142	Number of trend items with international parameters for the trend IRT scales in the FLA Parent Questionnaire
Table 22.A.143	Items in the FENGSP scale
Table 22.A.144	Items in the PARSAT scale
Table 22.A.145	Items in the PARIMPO scale
Table 22.A.146	Items in the PARENGL scale
Table 22.A.147	Items in the PARUSE scale
Table 22.A.148	Variables derived from the FLA Teacher Questionnaire (FLA-specific indices only)
Table 22.A.149	Cronbach's alpha for the IRT scales in the FLA Teacher Questionnaire
Table 22.A.150	Number of items with international parameters for the IRT scales in the FLA Teacher Questionnaire
Table 22.A.151	Number of trend items with international parameters for the trend IRT scales in the FLA Teacher Questionnaire
Table 22.A.152	Countries/economies for which the scale scores were suppressed for the IRT scales in the FLA Teacher Questionnaire
Table 22.A.153	Items in the JOBSATENG scale
Table 22.A.154	Items in the TCSELFATT scale
Table 22.A.155	Items in the TCSELFUSE scale
Table 22.A.156	Items in the TCUSEENG scale
Table 22.A.157	Items in the TCENGFORM scale
Table 22.A.158	Questionnaire Trend Items Parameters